

REVIEW

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A comprehensive review of the changing landscape of academic dishonesty in automated proctoring in the era of artificial intelligence

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Abstract

Exams are inherent in building confidence and appraising an individual in schools and colleges. However, supervising an exam is a tedious process given the large number of students in a class. This constant supervision is necessary to prevent students from cheating via copying, browsing, talking, or electronic means, thereby harming the integrity and authenticity of the examination. Technology has revolutionized how proctoring is undertaken, making this process more efficient by transitioning to online proctoring systems from traditional ones. Due to COVID-19, this online proctoring system has significantly shifted to utilize technologies such as Artificial Intelligence (AI) and Machine Learning (ML). Existing surveys focus on behavioral, pedagogical, and governance levels; however, there is a lack of discussion on applications, technologies, and associated challenges. This paper analyzes the challenges associated with conventional, online, automated, and AI-based proctoring systems and identifies the research gaps based on different applications and their impact on education and academia. Moreover, the paper provides conceptual scenarios related to automated online proctoring systems, insights into various categories, and the challenges associated with the survey, which will act as a boon to future researchers.

Keywords Proctoring systems, COVID-19, Online proctoring, Automated proctoring, Higher educational institutions

1 Introduction

In the early days, teaching and learning were guided and conducted through textbooks, paper, and pens [1]. Due to the invention of computers, education has shifted from conventional teaching methodology to PowerPoint presentations (PPT), videos, audio, and smart-board classrooms [2]. With the introduction of the Internet and ICT, huge demand and gaps related to skills development have been made easier by getting access to various technical resources across universities throughout the globe [3]. This phenomenon has shifted Higher Education Institutions (HEIs) from conventional teaching



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to online learning styles [4]. Online learning has brought access to various technical resources available to students through Massive Open Online Courses (MOOCs) [5]. In developing and underdeveloped countries, accommodating more students in a single class has eased the strain on HEIs to build separate physical infrastructure [6]. Additionally, it reduces the pressure on students to attend classes physically, especially considering the challenges of scheduling geographical restrictions. MOOCs address this problem by eliminating such restrictions by allowing students to enroll, learn, and earn a degree on online platforms from anywhere in the world [7]. MOOCs have also provided teachers with various multimedia resources for learning and teaching. According to a survey by Liu et al. [8], the percentage of students taking MOOCs increased to 151% between 2008 and 2018. While assessments of learning outcomes through exams are important in educational institutions, maintaining their authenticity and academic integrity is difficult, which is largely the responsibility of the proctor or the invigilator [9]. Conducting exams in schools and universities is challenging as it demands financial resources, infrastructure, verified proctors, and a cautious agenda. Examination venues are to be provided security, which is the major problem faced by current educational institutions when using offline proctoring methods [10]. Studies have found that nearly 37.8% of students (male and female) in college and 41.8% in high schools (both boys and girls) admit to being involved in cheating and misconduct during exams [11]. Therefore, monitoring students is essential, but it can be costly in terms of test center allocation and human resources to ensure academic fairness. This traditional method of monitoring students by means of a physical proctor is called offline proctoring, and its implementation in MOOCs may or may not require proctors, depending on the type of software used. The scenario of traditional offline proctoring has changed due to the spread of COVID-19. Due to sudden shift from traditional offline proctoring systems to online proctoring systems [12], this revolution has significantly impacted on the teaching and learning process, requiring students from high schools to HEIs to adapt to this new normal. Entrance tests, assessments, and every aspect of teaching and learning activity need to adjust, posing challenges at all levels.

Academic dishonesty in online learning has become so prevalent in the form of cheating, copying, and plagiarism [13] that it is next to impossible to ensure 100% integrity in the case of online learning, which led to the development of online proctoring. In online proctoring, the exam is monitored by a human proctor, either through a webcam or from a control center. This helps the authorities, staff, and mentor professionals to ensure academic integrity during the exams. Even in the case of online MOOCs, the mentors cannot guarantee whether the student has acquired the required knowledge and course materials. Though online proctoring is handy, it also has its disadvantages. The system can cause anxiety in test takers as it may involve users who are not fully proficient with the use of online tools. These factors can further contribute to situations that may lead to malpractice. Stressed students may choose unethical paths during exams by either cheating or doing anything that will ensure they perform better.

On the other hand, lack of proficiency amongst both the test takers and the proctors can lead to errors or misunderstandings during the monitoring process, thus leading to accusations of malpractice, even if that was not the case [14]. In addition, online proctoring requires infrastructure like webcams, microphones, and the Internet, which might not be affordable to all or readily available for students to have on their laptops. An

adequate number of staff or teachers are also needed to ensure that exams are conducted fairly. Online proctoring is essential to ensure academic integrity, combat cheating, provide accessibility in remote locations, and reduce logistical and infrastructure challenges. Many schools and universities rely upon publicly available proctoring platforms such as ProctorU or Kryterion or online testing services such as Speed Test, Examinations Online, and Think Examinations [14, 15]. This flexibility allows institutions to offer students the ability to take exams from anywhere. However, there are numerous challenges associated with online proctoring systems.

In offline proctoring, there is a risk of students engaging in dishonest activities in the absence of a proctor. Similarly, in online proctoring, a student has several opportunities to get involved in cheating. Though online proctoring assists us with a no-proctor facility, it has several disadvantages. A student may mask a person's face by simulating facial features via photographs or even videos, hence engaging in spoofing by copying the original identity of the other person. Additionally, they can use browsers and new tabs to search online while the exam is on, and they may copy and paste answers from books, mobile phones, or cheat sheets [16]. Online proctoring platforms generally require strong security and authentication mechanisms to address such challenges. However, in large exam settings, the efficiency of online proctoring systems is hindered by the inability of teachers to invigilate properly. Additionally, students may attempt to communicate with others or have someone else take the exam for them. Some institutions invest a hefty amount in the infrastructure and technology required for proctoring, which captures students' facial expressions and emotions through imaging systems. These systems often need persistent internet connectivity along with authentication, storage facilities, and security, which incur huge costs. However, examining all the video feeds or images involves a lot of computational time. To achieve an effective online proctoring system, a perfectly automated, artificial intelligence-based smart proctoring system, along with security, can provide a great deal in ensuring the academic integrity of the institutions.

1.1 Related works

Many surveys address different topics in online proctoring. Aurella et al. [17] investigated different proctoring methods and their future trends based on AI. Potluri et al. [18] surveyed fully automated online proctoring systems to detect the anomalous behavior of the students based on AI. Nigam et al. [19] focused on systematically reviewing AI and non-AI-based proctoring systems. Han et al. [20] have developed a systematic review of digital proctoring in HEIs. Henderson et al. [21] provided a systematic review of online examination proctoring based on scalable authentication and integrity. However, most of these surveys focus only on the types, technological advances, impacts, and limitations and do not focus on security, methodologies, conceptual scenarios, and insights that can improve the proctoring process in schools and HEIs.

1.2 Current survey

Though there is a wide variety of commercially available systems, none of the existing literature provides light on adapted technologies and their implications for a variety of applications and features for proctoring. This survey is intended to survey different methods, methodologies, and applications, along with a comparative analysis of various proctoring systems. To distinguish our survey from existing ones, Table 1 provides

Table 1 Comparison of this survey and other surveys

Survey and year	Aim	Methods	Applications	Security	Tools	Case study	Re-search issues
[17] & 2023	Ensuring exam integrity and safe administration in automated exam proctoring systems	Yes	Yes	No	Yes	No	Yes
[18] & 2022	In-depth analyses of AI based algorithms for examinee detection and recognition; Specific stress on examinee movements i.e. head pose and eye-gaze; examinee spoofing and gadget finding	Yes	Yes	Yes	No	No	Yes
[19] & 2021	Past, Present and Future giant leaps of AIPS are discussed; Parameters to be considered, trends, Issues and Future of AIPS	No	Yes	No	Yes	No	No
[20] & 2023	Digital Proctoring in HEIs	No	No	Yes	No	No	Yes
[21] & 2020	Various issues pertaining to online examinations	No	Yes	No	Yes	No	Yes
Proposed	Various issues pertaining to Online proctoring systems	Yes	Yes	Yes	Yes	Yes	Yes

a summary comparison of various objectives, such as aim, methods, applications, security, tools, conceptual scenarios, and research issues. Our survey addresses the gap by focusing on the applications, technology underlying these applications, functionalities, security, conceptual scenarios, and the latest and future advances in online proctoring systems. A comparative analysis focusing on systems, methods, technologies, conceptual scenarios, and security in online proctoring systems has been presented. This allows the researchers to better understand the trends in online proctoring.

1.3 Major contributions

Advancements in technology and the emergence of COVID-19 have had a significant impact on teaching and learning activities. This has led to a shift towards integrating various technical solutions employing artificial intelligence, deep learning, machine learning, web-related frameworks, and software products. However, cheating in online exams, maintaining academic integrity, and performance assessments become challenging for schools and HEIs. This section highlights the key factors constituting the proposed comprehensive proctoring system analysis. The primary contributions are as follows:

- A detailed survey has been conducted to examine proctoring systems, categorizing and classifying them based on their necessity, significance, and functionality and outlining their issues, advantages, and limitations.
- The classification and comparison are based on factors such as student cheating behavior, user activity recognition, proctoring methods, technologies, security, authentication, and more.

- An analysis of different technologies catering to various applications in proctoring has been provided, along with conceptual scenarios to showcase the various contributions of proctoring systems.
- In-depth analyses of conceptual scenarios and commercially available proctoring systems.
- Finally, research gaps and insights have been identified and proposed to address the needs of the future.

1.4 Structure of the paper

The paper has been organized as follows: Sect. 1 briefs about the introduction, existing surveys, and major contributions; Sect. 2 details the materials and methods involved for the design of the survey; Sect. 3 provides the classification of various kinds of proctoring systems and their functionalities, datasets, features, technology and applications; Sect. 4 provides the comparative analysis on security and performance metrics used in proctoring applications; Sect. 5 provides the comparative analysis on various commercially available proctoring systems and few conceptual scenarios; Sect. 6 provides the research insights pertaining to various proctoring applications; Sect. 7 provides the future research directions; and Sect. 8 concludes the paper. The acronyms mentioned in this survey are highlighted in Table 2.

2 Materials and methods

The methodology involves a search strategy, literature selection based on the PRISMA approach, identification of the research gap, and formulation of research questions. The survey was carried out after a careful understanding of the proctoring systems based on their applications in the fields of education and academia. A wide variety of literature pertaining to factors such as proctoring methods, technologies, security, and applications has been carefully selected to conduct the survey. Existing literature surveys [17, 18, 19, 20, 21] focused only on cheating behavior identification, automated proctoring, AI-based proctoring, and online proctoring systems. However, there is a limited focus on various types of applications in proctoring. Hence, the major focus has been on the applications, technologies involved, security, factors influencing these applications, and their benefits and limitations.

2.1 Search strategy

The search strategy involves the careful selection of literature based on the keywords. The keywords involve the use of keywords such as “AI proctoring”, “Online Proctoring”, “Offline Proctoring”, “Proctoring Applications”, “Machine Learning in Proctoring”, “Deep Learning in Proctoring”, “Security in proctoring”, “Authentication in Proctoring”, “Fully Automated”, “Semi-Automated”, “Automated”, “Digital Proctoring”, “live Proctoring”, “Key functionalities”, “e-Examinations”, “BYOD”. The search and the selection of the literature involve the research works and surveys from the commercially available paid and open-access databases such as IEEE, Science Direct, SCOPUS, and SCI databases.

2.2 Inclusion and exclusion criteria

Research and conference papers from Springer, EBSCO, ProQuest, Taylor and Francis, and other open-source databases were searched and extracted. The search spans from

Table 2 List of abbreviations

Acronym	Description
COVID-19; VGG	Corona Virus Disease-2019; Visual Geometry Group
MPIL	Max Planck Institute Informatik
FCNN	Fully Convolutional Network
AI; 3D; 2D	Artificial Intelligence; Three-Dimension; Two-Dimension;
LDA	Linear Discriminant Analysis
MATLAB; OFF	Matrix Laboratory; Optimal Feature Fusion
DL; VFOA;	Deep Learning; Visual Focus of Attention
LBPH; KNN	Local Binary Pattern Histograms; K-Nearest Neighbor
MWO	Modified Wolf Optimization
MOOCs; AdaBoost	Massive Open Online Courses; Adaptive Boosting
KTH	Kungliga Tekniska högskolan Dataset
CRNN	Convolutional Recurrent Neural Networks
ICT	Information and Communication Technology
CAVIAR	Computer Assisted Visual Interactive Recognition
GANs	Generative Adversarial Networks
HDBM	Human Motion Database
LCNN	Look-up-based Convolutional Neural Networks
HEI	Higher Educational Institutions;
AIDER	Aerial Image Database for Emergency Response Applications
IXMAS	INRIA Xmas Motion Acquisition Sequences
PPT; HSAR	PowerPoint Presentation; Human Suspicious Activity Recognition
NasNet; ORL	Neural Architecture Search Network; Olivetti Research Laboratory
ML; 3DCNN	Machine Learning; Three Dimensional Convolutional Neural Network
UCF	Un-constrained Face Dataset
C3D	Convolutional 3-Dimesional
IoT; FASSEG; Dlib	Internet of Things; Face Semantic Segmentation; Digital Library
CELEBA; LMS	Large-scale Celeb Faces Attributes; Learning Management System
HD; RF	High Definition; Radio Frequency Dataset
SEB	Safe Exam Browser
RP; ER; VR	Remote Proctoring; Extended Reality; Virtual Reality
AR; MR; GUI	Augmented Reality; Mixed Reality; Graphical User Interface
MitM	Man-in-the-Middle
AP; SQL	Automated Proctoring; Structured Query Language
IaaS; SaaS; DaaS	Infrastructure-as-a-Service; Software-as-a-Service; Desktop-as-a-Service
LP; HaaS; ANOVA	Live Proctoring; Hardware-as-a-Service; Analysis of Variance
NS3; GoCV; DoS	Network Simulator; Golang Computer Vision; Denial-of-Service
VP; APSE	Video Proctoring; Asymmetric Scalar Product Preserving Encryption
API	Application Programming Interface
CUI	Controlled Unclassified Information
SAP; GI4E	Semi-Automated Proctoring; Gaze Interaction for Everybody
COCO; GDPR	Common Objects in Context dataset; General Data Protection Regulation
SLAs	Service Level Agreement
NTU RGB-D	Nanyang Technological University's Red Blue Green and Depth info
CNN; ANN	Convolutional Neural Network; Artificial Neural Network
SVM; SLPs	Support Vector Machine; Service Level Policies
CPA	Certified Public Accountant
DCNN	Deep Convolutional Neural Network
CIA	Confidentiality, Integrity and Availability
ClaaS	Cloud-based Infrastructure-as-a-Service
HCI; DMCA	Human-Computer Interaction; Digital Millennium Copyright Act
VPN; IT; VMs	Virtual Private Networks; Information Technology; Virtual Machines
AAM; ACLs; WAPs	Active Appearance Model; Access Control Lists; Wireless Access Points
MAC; NLP	Medium Access Control; Natural Language Processing

Table 2 (continued)

Acronym	Description
OMR; AUC; G-Mean	Optical Mark Recognition; Area Under the Curve; Geometric Mean
UI; ULE	User Interface; Ubiquitous Learning Environment
OCR; TPR; TNR	Optical Character Recognition; True Positive Rate; True Negative Rate
MTCNN; 5G	Multi-Task Cascaded Convolutional Neural Network; Fifth Generation
ROC; IQ	Receiver operating characteristic curve; Intelligent Quotient
FPR; FNR; CV	False Positive Rate; False Negative Rate; Computer Vision
SELI	Smart Ecosystem for Learning and Inclusion
CTC; BOLL	Connectionist Temporal Classification; Blockchain of Learning Logs
KDE; RPA	Kernel Density Image; Robotic Process Automation;
PaaS	Platform-as-a-Service;
CaaS; CPU	Cloud-as-a-Service; Central Processing Unit
VFOA; GPU	Visual Focus of Attention; Graphics Processing Unit
OST; CLM	Omni-Sensing Technology; Constrained Local Model
OCSVM	One Class Support Vector Machine
GAVAM	Generalized Adaptive View-based Appearance Model
FCBF; AOI	Fast Correlation-based Filter; Areas of Interest
SMI	Senso Motoric Instruments
MCQ; ResNet	Multiple Choice Questions; Residual Networks
LSTM; KDE	Long Short-Term Memory Networks; Kernel Density Estimate
OpenCV	Open Source Computer Vision Library
I-HMM	Intelligent Hidden Markov Models
I-DT	Intelligent-Dispersion Threshold Algorithm
OGAMA; CIPL	Open Gaze and Mouse Analyzer; Crime investigation, prevention Lab;
FSA-Net	Fine-Grained Structure Aggregation Network
DBSCAN	Density-based Spatial Clustering of Applications with
LBP; SSD	Local Binary Pattern; Single Shot Detector
SIR	Sequential Importance Resampling
FLW	Faces Labeled in the Wild
MAE	Mean Absolute Error
MFDNet	Matrix Fisher Distribution Head Pose Network
CCC; SVD	Cross Category Center; Singular Value Decomposition
MAEV	Mean Absolute Error of Vectors
NDGNet	Non-uniform Gaussian-Label Distribution Network
INRIA	Infrared Head Pose Estimation
AADL	Anisotropic Angle Distribution Learning Network
SGD	Stochastic Gradient Estimation
HPE; RGB	Head Pose Estimation; Red-Green-Blue
MAPE; DNN	Maximum Aposterior Estimation; Deep Neural Networks
YOLO	You Only Look Once
SMQT	Successive Mean Quantization Transform
SNoW; HoG	Sparse Network of Windows; Histogram of Gradients
FER	Facial Expression Recognition
3WPCA-MD	Three-level Wavelet Decomposition Principal Component Analysis with Mahalanobis Distance
OST	Omni-Sensing Technology

2008 to 2024 publications. The publications must be peer-reviewed and available in English. Papers pertaining to peer-reviewed journals, conferences, and conference proceedings are included. Articles that belong to the subjective areas of the review. Validation studies pertaining to proctoring examinations, survey papers pertaining to exam dishonesty, student conduct, academic integrity, and online cheating are included. Articles that belong to offline publications other than the English language, papers pertaining

to implementation policies, summaries, existing papers, and plagiarized contents are excluded from the review. The selection strategy using the PRISMA method is depicted in Fig. 1. Further, a thematic analysis was carried out for the survey from the selected literature to identify the area where the research gap can be identified, and research questions can be formulated.

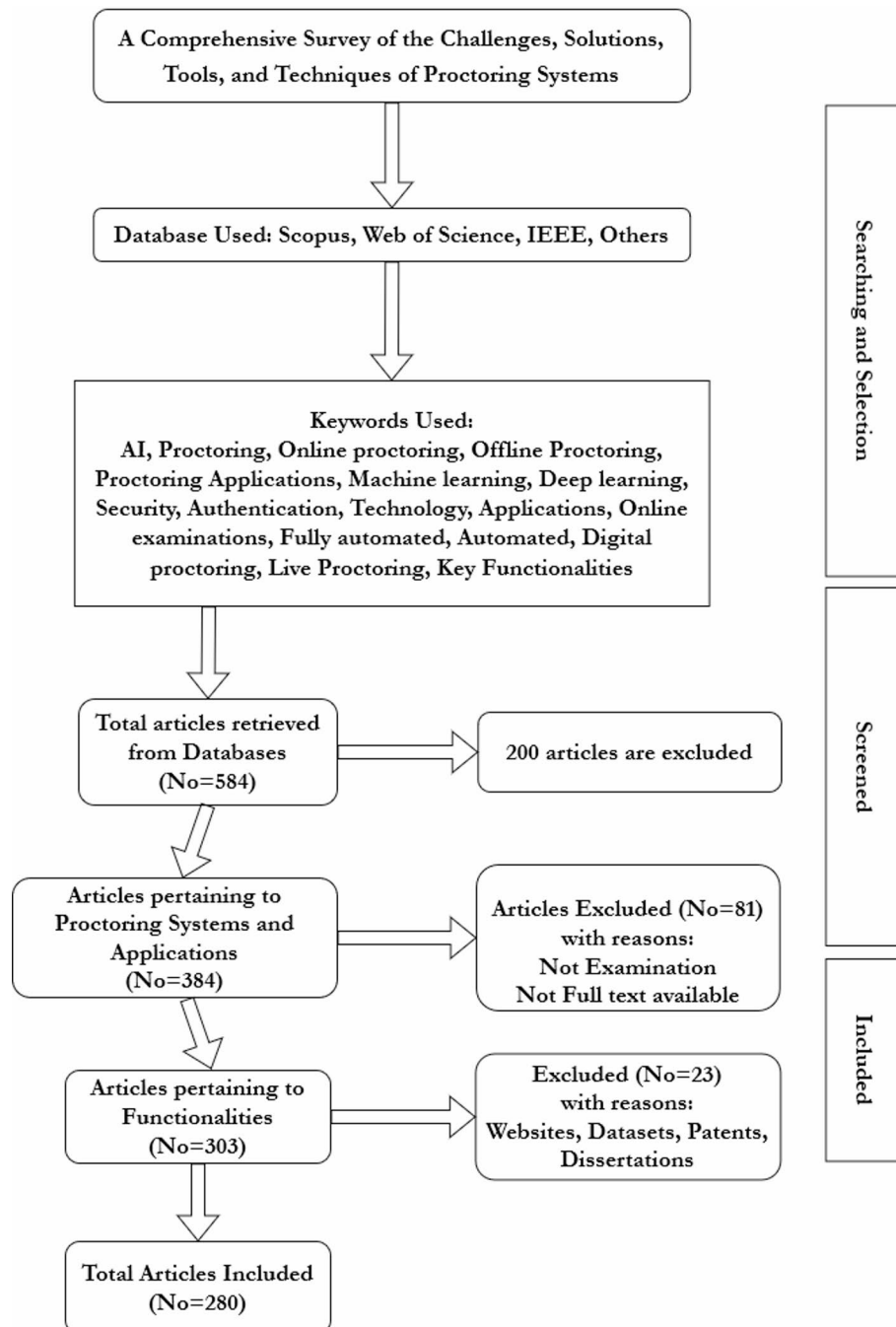


Fig. 1 Methodology of selection criteria

2.3 Identification of research gaps and formulation of research questions

The existing literatures haven't focused much in depth on different architectures, online proctoring issues, and their solutions. Therefore, a major gap has been identified in the area of proctoring applications, and this needs to be addressed. Further, this survey highlights the limitations pertaining to proctoring technology, the use of proctoring applications, and the challenges and countermeasures. The survey has been formulated as solutions for the following queries, which are coined as follows:

RQ1: What are the studies pertaining to proctoring solutions that involve cheating behavior detection?

RQ2: What are the technological implications pertaining to functionalities, features pertaining to online proctoring, offline proctoring solutions, and applications in the previous literature?

RQ3: What are the different ways in which the Cheating behavior detection of students can be understood?

RQ4: What are the security and authentication techniques utilized for proctoring and the implementation of proctoring solutions and applications?

RQ5: What are the challenges faced by different kinds of features engulfed in proctoring solutions and applications?

RQ6: What are the key aspects and major contributions made across proctoring technology and applications?

RQ7: What are the conceptual scenarios that highlight significant changes due to the pandemic and technological advancements in proctoring systems?

RQ8: What are the trends that contribute to the future of Proctoring Solutions?

2.4 Methodological limitations

While this systematic review provides comprehensive coverage of automated proctoring systems, there are several limitations of this study. First, the search was limited to specific databases, namely IEEE, Science Direct, SCOPUS, and SCI databases, which may have excluded relevant studies indexed elsewhere. Most importantly, the review focuses on peer-reviewed literature, however, only English-language publications were included, potentially excluding valuable research published in other languages, particularly from non-English-speaking regions, where online education is rapidly expanding. This is especially crucial as AI-based proctoring systems are trained on western behavioural norms, might not be trained on datasets from these regions, exacerbating cultural and social challenges. Further, studies with statistically significant or positive results are more likely to be published than those with null or negative findings. This may lead to overrepresentation of successful proctoring implementations while underreporting failures or ineffective systems. Despite these limitations, this review provides the most comprehensive synthesis of proctoring technologies, applications, challenges, and ethical considerations. The limitations identified here are inherent to systematic reviews and do not negate the value of the findings but rather contextualize them appropriately and identify directions for future research.

3 Classification of proctoring systems

Proctoring is defined as the process of supervising and managing examinations to ensure academic integrity and prevent cheating. Due to digitization and technological growth, various proctoring systems have been developed: offline proctoring, online proctoring. Further the online proctoring systems are categorized into live proctoring, semi-automated and automated proctoring systems [22, 23, 24]. Figure 2 represents the various types of proctoring systems identified from our literature study.

3.1 Offline proctoring systems

Offline proctoring, also known as manual proctoring requires the exam to be conducted at a physical location where the invigilator and the examinee must be present. This method can be stressful due to the need to arrive on time. Offline proctoring involves a proctor who physically verifies the presence of the examinee and their log details and looks out for cheating, tracking, and fraudulent activities. Despite the presence of a physical proctor, students tend to commit fraud and cheating. To ensure academic integrity, a competent proctor is necessary. Offline proctoring generally involves regular tests, computer-based objective-type tests, and Optical Mark Recognition (OMR) tests, requiring significant infrastructure and dedicated proctors, which can be costly. According to a survey with anecdotal evidence, students perform much better in online proctoring than in offline proctoring [25]. In offline proctoring, students have to pass through several access control mechanisms that pose greater stress for them. In addition, students performing offline exploit their knowledge and perform well because they are accustomed to performing the test on multiple occasions in the same venue. This limitation leads to failure in consideration of anxiety and fear, which are two critical factors that determine the self-efficacy of the examinees and affect their performance. However, counseling and guidance can boost confidence even in such a pressured environment.

3.2 Online proctoring systems

MOOC courses first came up with online proctoring systems. Online exams involve software fitted with a camera that keeps track of the student's behavior and prevents cheating. It allows students to appear for the exams at remote locations in their comfort

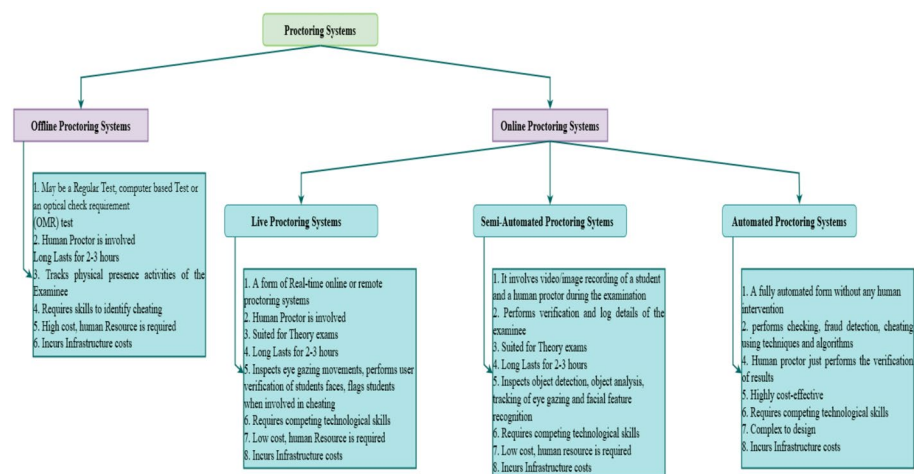


Fig. 2 Classification of proctoring system

[26]. Online proctoring systems commonly incorporate a camera-based monitoring component and a locking control, although the mode of camera usage may vary across live, automated, and semi-automated proctoring approaches. The former is used to record a video feed for reference of a delegated proctor to identify any mischievous behavior, fraud, or cheating committed by the examinee during the exam. The latter prevents the examinees from opening the new tabs in their browser window. This is also called a computer or browser lockdown, as proposed by Alessio et al. [27]. There are several other categories under online proctoring: live proctoring, semi-automated proctoring, and fully automated proctoring systems.

3.2.1 Live proctoring

In the case of medium-sized exams, live proctoring is the best option. Which is a combination of a virtual tool and a human proctor who would track and check in real-time with the help of an AI tool. The proctor will be at a distant location to check and track the test takers with the help of audio and video recordings with on-screening support. In case of any malpractice, misconduct, nodding the head, or looking weird, the proctors can chat with the examinee or terminate the exam within a stipulated period. To assist in this decision-making, AI tools run behind the scenes to assist the human proctor with evidence.

3.2.2 Semi-automated proctoring

This type of proctoring serves as an alternative to live proctoring. It involves both the video and image recordings of the entire test taken by the examinee. However, one drawback is the lack of internet connectivity and the infrastructure required for processing these images. Image-based proctoring can be valuable in addressing this issue. Under this method, the examinee's photos are taken at specified intervals: at the start of the exam, at the conclusion, and during the answering process, every 30 to 45 s. These images are used to ensure academic integrity and to monitor the candidate's behavior for any misconduct.

3.2.3 Fully automated proctoring

It is well suited for large-scale exams. It uses fully automated AI tools that roost over the host to detect eye gazing, facial features, nodding the head, tracking hand movements and user verification, and activity recognition to prevent any cheating. The software integrates progressive AI and data analytics tools to assess and generate a report at the end of the exam. The major advantages of fully automated proctoring are cost and time efficiency, improved flexibility and credibility, absence of human involvement, and a reasonable cheating-free environment, which can be monitored using the performance index.

3.3 Key features of online proctoring systems

When the examinee starts the exam in online proctoring, a separate software program must be installed on their systems. This software mostly includes a lockdown feature that prevents the browser from opening a new window to prevent cheating. Some key functionalities involved in online proctoring are eye gaze tracking, head pose estimation, facial recognition, user verification, activity recognition, and authentication. This section

categorizes different works related and illustrates the key features for research through Fig. 3.

3.3.1 Eye gaze tracking

Eye gaze estimation or tracking is a process of determining the motion of a human eye to gauge and analyze the gaze pattern using sensors and algorithms [28]. It provides a correlation between eye movement and the environment. They are utilized in various applications, such as human behavior and cognition, usability testing of user interfaces, and assistive technology for individuals with disabilities. Different technologies provide stimuli for eye gaze estimation, namely infrared eye tracking, and video-based eye tracking. Some key metrics utilized in eye gaze tracking are fixation, gaze points, heatmaps, Areas of Interest (AOI), and Fixation Sequences [29]. Some factors that govern eye gaze tracking systems are accuracy, portability, compatibility, and user experience [30]. This section provides an overview of some of the research proposed by various researchers about eye gaze estimation related to online proctoring systems.

Liu et al. [31] aimed to explore the AI-based visual proctoring approach by using surveillance cameras in real-world offline examination rooms to assist proctoring. This approach proposed a visual assessment system followed by the proposed 2-Longitudinal-Stream CNN under different light conditions. The study utilized a dimensionality reduction technique known as Mini-Batch cutting for video data to convert the video. The system is implemented using OpenCV-python, where the models are trained with a data augmentation model and fed to CNN. Their proposed model has achieved a higher detection rate compared to other conventional approaches. Yaqub et al. [32] have developed an online proctoring system that utilizes eye tracking to detect whether the examinee is looking at the computer screen by employing a face-hiding concept. Initially, the system verifies the identity of the examinee in three steps: face verification and OCR. Face verification involves using the MTCNN and FaceNet, and the results are processed through a Siamese network to enhance data representation. The cosine-similarity score is utilized to determine the examinee's authenticity. MTCNN is used for face detection, while EasyOCR is used for text detection and recognition using ResNet + LSTM + CTC, followed by eye gaze estimation with an anomaly detection module. The proposed model utilizes recorded videos to track eye gaze and detect any cheating behavior of the examinee.

Jagendra Singh et al. [33] have created a classification system for exam proctoring using eye gaze detection. Their system features an Eagle eye to detect eye gaze using AI and Machine learning techniques. The proposed system involves a video feed to extract

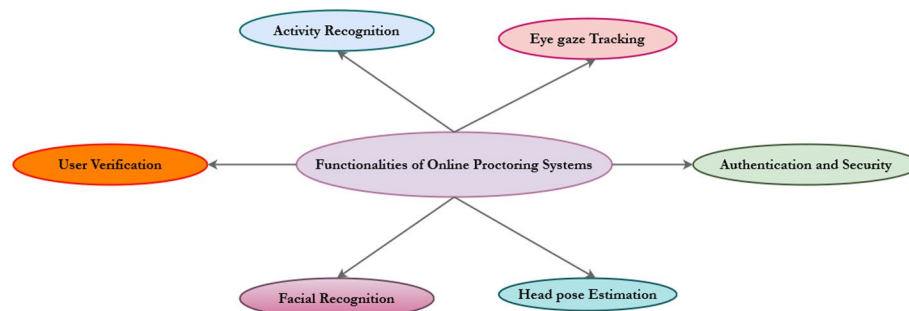


Fig. 3 Classification of Key Functionalities of Online Proctoring Systems

the frames, followed by eyeball localization. The eyeball localization is then classified using the Haar Cascade classifier, which was trained and tested using CNN. The model achieved an accuracy of more than 98%.

Li et al. [34] has introduced an automated online proctoring system that utilizes Gaze View Tracking. This system involves a 3D gaze motion formula and a cheating classification using a custom-trained object detection model. Their Gaze View Tracking system leverages two cameras and facial landmark detection. The output is then processed by an AI-based cheating classifier and a TensorFlow object detection algorithm with a custom-trained object detection model to identify pre-configured cheating targets. Their proposed model has achieved an impressive accuracy of 95%.

Dilini et al. [35] suggested a method for cheating detection in browser-based online exams using eye gaze tracking. They developed a browser plugin that tracks eye gaze movements captured using the built-in web camera. A dataset is generated from the eye gaze data while a student takes an online exam. They used a One-Class Support Vector Machine (OCSVM) for cheating detection and evaluated their model using Matplotlib and Seaborn KDE. The system can predict both cheating and non-cheating behaviors of the student, achieving a detection accuracy of 95%.

Indi et al. [36] presented a system for detecting malpractice in e-exams using head pose and gaze estimation models. Their system classifies the student's VFOA data using ML techniques. It processes video frames with Haar feature-based cascade classifiers and a pre-trained FSA-Net Model, obtaining an accuracy of 96.09%.

Lim et al. [37] conducted a study on gaze analysis using a webcam eye-tracking module. Their system has the advantage of allowing remote testing, which does not require any special hardware. The study focused on the gaze cueing effect on 52 participants. They used the EYEVIDEO lab, cloud software for webcam eye tracking analysis, and measured performance using metrics such as AOIs, mouse data, and gaze plots. Their analysis shows that gaze data is not related to the stimulus but to the gaze path to gauge overall attention toward orientation.

Ramachandra et al. [38] have developed an intelligent Eye Gaze-based assessment system to gain deeper insights into the performance of the learners. The system uses webcam-based tracking to assess learners in terms of performance metrics such as engagement, attention to critical information, wavering behavior, knowledge acquisition, and confusion. They employed MCQs designed using the SMI experiment suite 360°. The I-HMM and I-DT algorithms are used to identify groups of gaze points, and pyGaze is utilized to capture raw eye data. The eye movements are captured using a remote video-based SMI Redn professional eye tracker developed and interfaced with the pyGaze Analyzer designed in Python 3.7. Their proposed system provides personalized feedback for learners about different factors that affect their learning capability. However, their limitation is that they provide personalized reports on independent tasks and do not continuously monitor learners' capabilities. Another limitation is that the designed system works well only with objective-type assessments and cannot focus on subjective assessments. Additionally, it does not provide any insights into cognitive parameters related to higher-level thinking of Bloom's taxonomy. Similarly, the data analysis included only eye gaze data and a limited number of participants.

Ujbanyi et al. [39] performed an eye-tracking analysis by using a pilot application for a computer exam test. Their system analyzed human behavior and cognitive parameters

by tracking eye movements. They employed specialized hardware called Eye Tribe Tracker, a camera-based eye tracker tool that measures two types of eye movements: fixation and saccadic motion. The system is interfaced with OGAMA open-source software and provides educators with real-time feedback on attention levels, direction, and focus. This data can be used to frame appropriate knowledge transfer methodologies. Though this system provides deeper insights, it poses limitations when multiple participants are involved in the experiment. Gopane et al. [40] introduced an AI-based online exam-enhancing monitoring system that detects micro-expressions such as laughter detection, eye gaze tracking, eye blinking/closing, and head activity/head movement direction. Their proposed system uses video feeds to identify head movements through the Lucas Kanade algorithm, track eye gaze using the Active Appearance Model (AAM), and detect micro-expressions using DCNN, which are then classified using SVM. Their FaceNet model involves face detection using a Multi-task Cascaded Convolutional Networks (MTCNN) with a precision of 100%.

3.3.2 Summary and insights

In the case of online proctoring, eye gaze estimation and tracking play a prominent role in understanding the cognitive parameters of the learners. Additionally, it allows educators to frame their learning methodology pertaining to the student's understanding capability, focus, and direction. Most works focus on eye gaze estimation under varying light conditions. However, dim light conditions have to be considered. Eye gaze estimates are highly facilitated by the help of ANN for estimating eye gaze points or CNN for regression and prediction problems captured from the eye images [41–44]. Gaining a profound understanding of making decisions and solving problems is facilitated by eye gaze monitoring and detection [45]. Appearance-based gaze estimation requires the computation of visual saliency maps from the video clips, which are implemented via multimodal-CNN [46, 47]. The highlights mentioned involve applications pertaining to HCI [48], gaze points from oculography [49–51], and calibration methods [52, 53]. However, gaze estimation methods suffer from problems such as continuous monitoring, concentration on subjective assessments, and limited participants. In order to increase the accuracy of gaze estimation, the utilization of high-end cameras and sensors fitted on eye trackers to perform object detection, gaze points, and fixations is essential to determine the learning rate of the examinee. Transfer learning and other deep learning strategies need to be considered in order to facilitate live control and management towards protection against the cheating behavior of students in the examination. Table 3 comprehensively analyzes various online proctoring systems pertaining to eye gaze estimation and tracking. Figure 4 provides a generic framework for eye estimation, head pose estimation and facial recognition in Online Proctoring Systems.

3.3.3 Head pose estimation

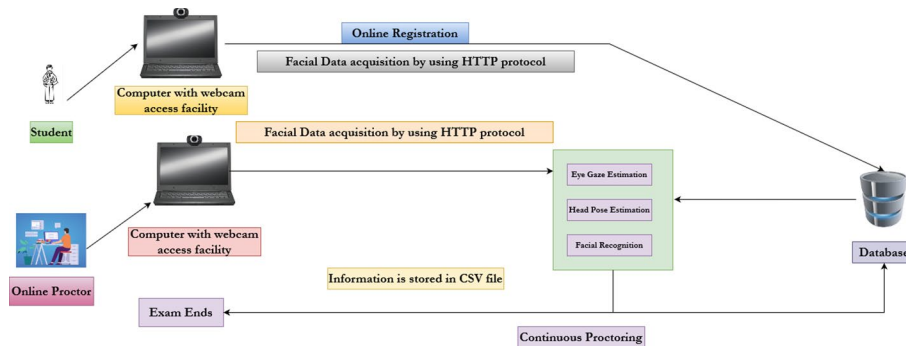
Head pose estimation is one of the key components or applications involved in identifying students' cheating behavior in online proctoring systems. Specifically, Yaw angles are utilized to detect malpractice and prevent cheating during the examination. Pose estimation proves to be a solid solution for detecting cheating in online exams [54]. There are two types of head pose estimation: single pose estimation and multiple pose estimation. This estimation is highly facilitated by the extraction of feature points. This

Table 3 Comprehensive analysis pertaining to eye gaze estimation/tracking

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Liu et al. [31]	AI-based Visual Assessment Proctoring System	2-Longitudinal-Stream CNN	Cheating Behavior Detection	Python 3.7, Keras 2.3, and Tensor-Flow 2.1 Data samples- 2700	Reduces the work-load and improves accuracy under varying light conditions	1. 3D-CNN can be considered to improve performance. 2. Accuracy needs to be improved by using stride and background subtraction algorithm
Yaqub et al. [32]	Proctoring-Eye Gaze Tracking	Face Verification-MTCNN; FaceNet OCR-ResNet + LSTM + CTC	Cheating Behavior Detection	Mediapipe and Dlib	dHash model performs much better on masked anonymized data	Accuracy needs to be improved further, which is very low when compared with its counterparts
Dilini et al. [35]	Eye Gaze Tracking	One class Support Vector Machine (OCSVM)	Cheating Behavior Detection	Matplotlib and Seaborn KDE	Achieved better accuracy in detecting cheating	The developed system has to be tested for different types of cheating. It runs with limited data size.
Ramachandra et al. [38]	Intelligent Eye Gaze based Assessment System	Intelligent-HMM; Intelligent-DTA	Cheating Behavior Detection	SMI Redn professional eye tracker 360 ⁰ ; PyGaze Analyzer; Python 3.7	Provides insights about their learning ability through personalized reports	1. Does not provide continuous monitoring of learners' capability 2. Cannot provide insights into cognitive parameters 3. Only gaze data and involves only a limited dataset.

Table 3 (continued)

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Ujbanyi et al. [39]	Eye Tracking Analysis	Eye Tribe Tracker	Cheating Behavior detection	OGAMA Open Source Software	Provides real-time feedback about the level of attention, direction, and focus of the learners	Number of participants involved is very limited.
Gopane et al. [40]	AI-based online exam enhancing monitoring system	Lucas kanade algorithm, AAM, DCNN, SVM, MTCNN	Laughter detection, eye gaze tracking, eye blinking/closing, and head activity/head movement direction.	Python	Achieved a precision of 100%	-

**Fig. 4** Generic framework for eye estimation, head pose estimation and facial recognition in online proctoring systems

section presents a comprehensive view of various studies pertaining to head pose estimation in online proctoring systems to prevent cheating. Potluri et al. [55] have developed an attentive system to address the issue of identifying malpractices by testing four components: face detection, multiple-person detection, face spoofing, and head pose estimation. The system uses Attentive Net to detect students' faces using the bounding box technique and aligns the faces using a rotation matrix of affine transformation. The landmarks and facial features are extracted using the FaceNet algorithm, which includes the Inception-ResNetV1 block. Face spoofing is detected using a shallow CNN Liveness Net, and the head pose estimation is achieved by solving the PNP equation. Their system utilized the CIPL and CASIA-Web Face datasets, achieving an accuracy of 87%.

Chuang et al. [56] investigated the probability of cheating during online exams by analyzing time delay and head pose estimation. Their proposed system identifies the relevant and non-repetitive VFOA characteristics associated with cheating and determines which features can predict cheating behaviors. Their system was developed using Python and analyzed using the CLM-GAVAM model. To decrease the complexity of the data, they introduced FCBF and achieved an accuracy of 75.6%. Irfan et al. [57] have developed an

online proctoring system to ensure academic integrity during online examinations by detecting abnormal student behavior. Their proposed system focused on processing the video and audio feeds from the webcam. It detects abnormal behavior in the candidate by estimating the head utilizing three angles: yaw, pitch, and roll. The system takes periodic snapshots using the webcam based on available bandwidth and time intervals and utilizes FSA-Net for anomaly detection, DBSCAN for outlier detection, and One-class SVM for head pose prediction. Audio feeds are used to detect the examinee's voice for signs of cheating. The system uses LBP, MTCNN, SSD, and OCSVM for face detection and employs supervised learning. The key advantage of their model is the utilization of FSA-Net to detect the face of the human behind the test taker. However, their model needs to be tested in various real-time online test scenarios.

Cote et al. [58] introduced a model for identifying abnormal behavior in remote online examinations. This model produces video summaries during the exam or post-exam reviews. The methodology uses a two-state hidden Markov model to identify the examinee's normal and abnormal behavior based on their head pose estimation. Sequential Importance Resampling (SIR) is used to track head movement and is implemented as the particle filter algorithm. The methodology was implemented using Python with OpenCV for behavior modeling, detection, and video summarization, while Matlab was used for video dataset handling. The proposed system can be readily utilized for live assistance by generating warning alerts as the model identifies the abnormal state. Though their model achieves feasibility, it cannot be compared with any other alternative methods. The methodology has to be tested under complex conditions, and the integration of audio analysis could also be explored.

Liu et al. [59] presented a head pose estimation model based on ARHPE in industrial HCI. Their methodology involves two main objectives. Firstly, they used a 2-dimensional Lorentz distribution to reconstruct all the incorrect labels of head pose images. Secondly, they employed asymmetric relation techniques to learn the adjacent poses. Their proposed model consists of a DeepNN comprising three modules: ResNet-50 backbone structure, 2-D Lorentz distribution learning for similarity mapping among adjacent poses, and regularization using Jeffrey's (JFK's) divergence. Their methodology utilizes the Pointing '04, FLW dataset, and HRIHP implemented with the TensorFlow toolbox using an Adam optimizer. The effectiveness of the model is assessed using Accuracy and Mean Absolute Error (MAE). The proposed model effectively addresses the conflict between the performance of the prediction model and incorrect labels. However, they suggest that incorporating depth and temporal information could further improve the accuracy of the head pose estimation.

Badave et al. [60] proposed a face recognition model for head pose estimation using a robust multi-camera system. Their system uses video feeds and face landmark models from the Dlib library to track facial features. Their approach achieved 100% accuracy in estimating the yaw angle variation of head poses. However, the model requires testing under different test scenarios relating to a variety of illumination conditions, occlusions, and head posture variations. Zhang et al. [61] developed an FDN for head pose estimation. Their approach aimed to tackle the problem of losing spatial information and dealing with large head pose variations. They proposed a three-branch network architecture called the FND for landmark-free head pose estimation from a single RGB image to address this. They tested and implemented their methodology using the 300 W-LP

dataset, AFLW2000 dataset, and BIWI datasets. In their work, they used a feature decoupling module to separate the hybrid features and utilized Cross-Category Center (CCC) loss to adjust the differential features by altering the channel responses. While their model performed well, detailed information about its implementation is lacking.

Liu et al. [62] tackled the challenges of dealing with ambiguity and disturbances during head pose estimation by developing a robust three-branch model by incorporating a triplet module and matrix Fisher distribution module. The proposed approach estimated the head rotation by employing an exponential probability density model based on the rotation matrix. The model utilized the machine learning technique SVD to achieve prediction and high accuracy and was implemented using 300 W-LP, AFLW2000, and BIWI datasets. Video feeds are trained using the MTCNN face detector and coded using PyTorch. The proposed MFDNet model effectively mitigates disturbances such as identity, illumination, and occlusion, resulting in an original head pose with a reduced error rate. Cao et al. [63] addressed the challenge of dealing with annotations that have discontinuities and errors in head pose estimation. They devised an annotation method using three vectors and assessed its performance using Mean Absolute Error of Vectors (MAEV) on datasets such as AFLW2000, BIWI, and 300 W-LP under MTCNN. Though their module demonstrated a low error rate of MAE, it still requires further testing for different test scenarios.

Liu et al. [64] shed light on the problem associated with low-quality IR images and ambiguity in head pose estimation by introducing a novel Non-uniform Gaussian-label Distribution Learning Network (NGDNet). They tested their methodology using the IRHPE, CAS-PEAL-R1, and Pointing'04 datasets and measured the performance using metrics such as MAE and MA. Their model achieved an accuracy of 99.31% on the Pointing'04 dataset under yaw angle variation. The proposed methodology's performance has been optimized using a mini-batch gradient descent algorithm, achieving high accuracy under weak illumination conditions. However, the model has not achieved optimized performance under diverse head pose angle variations. Liu et al. [65] presented the Anisotropic Angle Distribution Learning Network (AADL) model for head pose estimation to understand human attention in HCI. Their systems utilized maximum posterior technology to construct the head pose estimation network, which consisted of a convolutional layer, a covariance pooling layer, and an output layer. The prediction layer was prepared using Kullback-Leibler divergence, which was implemented with CNN. However, the proposed methodology does not account for missing angles and blurriness in head pose estimation and requires implementation using video feeds.

3.3.4 Summary and insights

In online proctoring, head pose estimation plays a major role in determining whether the examinee is within the frame of a video camera or not. It largely involves feature extraction via the Yaw, Pitch, and Roll angles. However, other head pose positions with their diverse angles can also be taken into consideration, and temporal information can be incorporated into head pose estimation. Optimization techniques that concentrate on these other angles can help in assessing the integrity of the student, especially under challenging and varying dim light conditions. Estimating blur and identifying missing angles are areas that are yet to be addressed to predict the students' attentiveness during examinations.

Using diverse datasets that cover varying illumination, occlusion, and video and audio feeds can enhance accuracy and classification. Researchers must also work on constructing risk estimation models to estimate each type of suspected behavior effectively. Head poses such as fast disappearance and abnormal head poses may also be investigated [66]. Table 4 provides a comprehensive analysis of various online proctoring systems pertaining to head pose estimation.

3.3.5 Facial recognition

Facial recognition typically involves identifying and measuring facial features in an image to confirm a person's identity [67]. Proctoring is identifying a student's cheating behavior using facial features through audio or video feeds. AI-based proctoring assists in identifying unusual motions or behaviors of exam takers that may indicate cheating and confirms their identity to prevent impersonation [68]. There is a fundamental difference between face detection and recognition. Face detection locates human faces in images using AI technology, whereas recognition involves identifying the person's identity. Face recognition has been accomplished using models such as Haar, dlib, Multi-task Cascaded Convolutional Neural Network (MTCNN), and OpenCV's DNN [69]. This section provides a comprehensive explanation of various research works on facial recognition in AI-based proctoring systems.

Elharrouss et al. [70] established an MTCNN model to recognize faces under different head pose variations. They utilized CNN for face recognition with the FLW, CEP, and CASIA-WebFace datasets and evaluated the model using the Pointing04 and Schneiderman datasets. Their model achieved an accuracy of 99.21%. However, testing different neural network recognition models could improve their accuracy. Kulshrestha et al. [71] proposed an AI-based exam proctoring system capable of gaze detection, facial recognition, multiple-person detection, and mobile/object detection. Their system relies on CNN-based object detection to identify cheating in online exams. They employed the MTCNN library for face detection, covering multiple-person detection, mobile/object detection, and eye gaze tracking. However, their system lacks performance investigation and experimental analysis.

Ganidisastra et al. [72] proposed visual verification for online exams using a CNN-FR approach. They suggested using SURF for image equalization and feature extraction and the incorporation of an incremental training approach for face recognition to reduce computational time and cost. In their proposed system, facial recognition occurs through face detection, feature extraction, and facial verification, and the Viola-Jones/Haar Cascade Method, Local Binary Pattern (LBP), MTCNN, and YOLO face are used to accomplish the task. However, their method has limitations in dim-light conditions and blurring, among other issues. YOLO Face achieved an accuracy of 98.22%, and MTCNN achieved 98.63%. The report also concluded that it is essential to focus on implementing face detection and recognition in smartphone environments in the near future.

Sasikala et al. [73] created a facial recognition-based automated online proctoring system for administering online exams. Their primary objective was to detect any misconduct during the online exams and monitor activities such as tab switching, opening multiple browser windows, using webcams and mobile devices, and the presence of numerous individuals. Their study utilized the Viola-Jones face detection algorithm involving Haar Feature Selection, Successive Mean Quantization Transform (SMQT)

Table 4 Comprehensive analysis pertaining to head pose estimation

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Potluri et al. [55]	AI-based proctoring-Attentive System	Attentive-Net	Misbehavior detection	Video Feeds 80×80 frames Keras with TensorFlow	Cheating Behaviors are identified with an accuracy of 86%	1. Partial or Occluded faces are not properly identified 2. Examinees with a neutral expression are also identified as misbehavior
Chuang et al. [56]	Cheating detection based on time delay and head pose	CLM-GAVAM Algorithm; Fast Correlation-based Filter	Cheating Detection	Python	1. Accuracy is 75.6% under time delay 2. Reduced the expected time delay to discover infractions on recorded exams	Does not indicate whether a student has actually cheated. Instead, it shows whether a student is behaving in a manner that is consistent with cheating
Liu et al. [59]	Asymmetric Relation Aware Representation Learning-HPE	2-D Lorentz Distribution Function-DEEPNN; Multivariate Label Distribution	Head Pose Estimation	Tensor Flow Tool Box	Under pitch angle, 95.47% accuracy, and at yaw angle, 95.16% accuracy	Depth/Temporal information has not been considered
Zhang et al. [61]	Landmark-free HPE for RGB Images	Feature De-coupling Network	Head Pose Estimation	ImageNet; SGD Optimizer; PyTorch	Pixel intensity information is used to minimize the error by 9.4% on the BIWI dataset	Implementation is complex
Liu et al. [62]	Collaborative pose estimation	Matrix Fisher Distribution Network (MFDNet) Triplet Architecture	Head Pose Estimation	PyTorch	Avoids Ambiguity problem Robustness is improved	-
Liu et al. [64]	On-Task Behavior understanding	Non-uniform Gaussian-label distribution learning (NGDNet); Mini-batch Gradient Descent Algorithm; Kullback-Leibler Divergence	Head Pose Estimation	-	1. Accuracy is 99.08% on the IRHPE dataset and 87.41% on CAS-PEAL-R1 datasets 2. Beneficial to the HCI in a weak illumination condition	1. Lags implementation details 2. Video sequences have to be decided

Table 4 (continued)

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Liu et al. [65]	Understanding HCI	Anisotropic Angle Distribution Learning (AADL)-CNN network; Kullback-Leibler Divergence; Maximum a Posterior (MAP) estimation method	Head Pose Estimation	TensorFlow	Missing Angle and Motion Blur conditions are also considered in their work and appear to achieve high performance	Head Roll directions are not yet considered.

features, and Sparse Network of Windows (SNoW) classifier methods. Face detection uses MTCNN-FaceNet, while classification uses the SVM. However, their proposed methodology lags in detail about implementation and experimental analysis. Ahmad et al. [74] formulated a deep learning-based online proctoring system for facial recognition. Their system monitors students to identify any unfair, unethical, and illegal behavior during classes and exams. They employ biometric approaches such as facial recognition using the Histogram of Gradients (HOG) face detector and the OpenCV facial recognition algorithm, achieving an accuracy of 99%. However, their system only incorporates the YOLOv3 detection model, and there is enormous scope for further research.

Ozdamli et al. [75] introduced a facial recognition system to detect student emotions and cheating behavior in distance learning. Their approach incorporates the 3WPCA-MD (Three-level Wavelet Decomposition Principal Component Analysis with Mahalanobis Distance) algorithm for emotion detection and utilizes PCA for feature extraction. The system achieved an accuracy rate of 98% in facial recognition and supports attendance tracking. It leverages data from the FER dataset and is implemented using Python 3.3.8. Emotion recognition is performed using the mini-Xception CNN model. However, the system currently faces challenges such as the inability to detect head and face movements automatically, difficulty in distinguishing the same emotion exhibited by the same student, and limited performance in areas with poor internet connectivity.

Sirivarshitha et al. [76] developed a face detection and recognition system using OpenCV and Face Recognition Libraries in Python. They transformed the mathematical concepts of a face into a face print and implemented the system using OpenCV and Python libraries. Additionally, they used a 3D pre-defined face alignment model for face detection. Their model successfully identified multiple faces with an accuracy of 80%. However, they aim to improve the accuracy of their methodology. Nurpeisova et al. [77] studied mathematical models and algorithms for face recognition in images using Python for proctoring systems. They developed a model that was implemented using Python in the OpenCV framework. The mathematical model was tested using CNN with a Haar-based feature selection technique, resulting in an accuracy rate of 98.95%. Though the system has achieved significant accuracy and reduction in identifying false positives, it suffers from problems such as overlapping faces, various head turns and tilts, changes in illumination intensity and angles, and recognizing different facial expressions. Additionally, errors are inevitable when using video sequences to evaluate the implementation of the proposed work.

Nirmala et al. [78] proposed an automated exam control and proctoring application using Deep Face Recognition. They utilized a VGG model in combination with CNN and implemented the system using Keras and TensorFlow, with the DeepFace library, Python, and JavaScript. The FACENET model and cosine similarity were used to classify feature selection. The accuracy rate was observed to increase with the number of epochs. Adhikary et al. [79] developed a facial emotion recognition model using real-time 2D pose estimation. Their methodology combined facial emotion recognition with body pose estimation to better determine a person's actions and behavior. They employed DeXpression network architecture and the inception blocks of GoogleNet to extract facial key points. Their network had 4 stacked FaceEmoteBlocks, predicting 7 different types of emotions. They achieved an accuracy of 99.3% using the MPII dataset, FCN-RESNET, and CNN for the study. Their model involved improving coordination extraction from heat maps by random selection rather than the maximum value, but there is still room for improvement in identifying body parts during recognition.

Geetha et al. [80] developed a face detection and recognition system using ML algorithms for online examinations. Their study aims to increase the accuracy of the facial recognition system by employing SVM and Eigenface algorithms. They found that the model accuracy increases as the size of the dataset is expanded. However, their model witness's implementation drawbacks, thereby reducing its overall efficacy. Yang et al. [81] proposed a novel exam monitoring and analysis system based on face detection and recognition. Their system assists invigilators in real-time student identification to detect abnormal behaviors via face recognition during the post-exam video analysis. The system has been implemented on the Zoom platform, where video streams are chosen as input and optical character recognition is utilized for detection. PCA and LDA are employed with the CNN model for face detection. Their system used OpenCV and JavaScript with a DNN face detector. Feature extraction involves the use of Haar Cascade Selection. The model has achieved an accuracy of 90% on ResNet152, but adjustments are needed for implementation in challenging environments.

3.3.6 Summary and insights

Most of the proctoring systems utilized either facial pose detection or emotion recognition based on angles such as the yaw, pitch, and roll axes. Additional unorganized angles can be considered to detect the facial poses and abnormal examinee behavior. While many CNN models have been proposed, limited work has been done on optimization and AI-based techniques. It is crucial to explore multiple face detection methods using landmark localization techniques [82]. The four main types of face detectors are Haar-Cascade, LBP, MTCNN, and Yolo-Face, and it's also worth exploring other algorithms, such as crucial point positioning technology [83]. Furthermore, 3D approaches to identifying faces in a crowd can help solve existing issues with 2D-based face recognition models [84]. Table 5 comprehensively analyzes various face recognition models used in online proctoring systems.

3.3.7 Activity recognition

Monitoring test-takers for potential cheating is a crucial aspect of proctoring systems. Detecting suspicious behavior requires prominent proof to uphold the integrity of the examination. To identify such dishonest activities during exams, computer vision can

Table 5 Comprehensive analysis of face recognition in online proctoring systems

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Elharrouss et al. [70]	Pose-Invariant Face Recognition in the wild	Multi-task Cascaded Convolutional Neural Network (MTCNN)	Recognition of faces using different pose variations	-	Achieved an accuracy of 99.21%	Model performance has to be tested and evaluated under various neural network models.
Kulshreshta et al. [71]	AI-based proctoring system	CNN	Proctoring	-	-	Lags in implementation and analysis
Ganidisatra et al. [72]	Online Exam proctoring	Haar, LBP, MTCNN, YOLOv3 Face	Incremental Training on Face Recognition	OpenCV, Python, TensorFlow	YOLO-Accuracy 98.22% MTCNN-Accuracy 98.63%	Face detection under smart mobile environments needs to be considered
Sasikala et al. [73]	Facial recognition based automated online proctoring system	Haar feature selection, Successive Mean Quantization Transform (SMQT) features and Sparse Network of Windows (SNoW) classifier	To identify the malpractice during exams	-	Detects the misbehaving student's face efficiently	Lags the implementation details and questions the integrity of the work.
Ahmad et al. [74]	Deep Learning based Online proctoring using face recognition	Histogram of Gradients (HoG), Eigenface, Fisherface, Local Binary Pattern Histograms (LBPH)	Facial Recognition Algorithms	Yolov3, OpenCV, Python, Dlib	Achieved an accuracy of 99%	It has to be tested in real-time environments
Ozdamli et al. [75]	Facial Recognition System	Three-level Wavelet Decomposition Principal Component Analysis with Mahalanobis Distance (3WPCA-MD), PCA-feature extraction	To identify emotion and facial expression	Python 3.3.8	Accuracy 98%	Inability to automatically detect head and face movements
Sirivaritha et al. [76]	Face detection and Recognition	Face print, Face Alignment model	To identify the misbehavior activity of students	OpenCV, Python	Accuracy 80%	Still, improvement is needed in terms of accuracy
Nurpeisoova et al. [77]	Online proctoring system	CNN, Haar-Based Feature Selection	To identify the misbehavior of the student	OpenCV, Python	Accuracy 98.95% Reduces the false positives to a considerable rate	Suffers from problems such as overlapping faces, various head turns and tilts, illumination intensity and angles, and facial expressions

Table 5 (continued)

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Adhikary et al. [79]	Facial Emotion Recognition Model	2D Pose estimation DeXpression Network, FCN-RESNET CNN	To identify the face poses	-	Accuracy 99.3%	Still, an improvement is needed in terms of identifying the body parts while performing recognition.
Yang et al. [81]	Exam Monitoring and Analysis system	PCA, LDA, CNN	To detect abnormal behaviors	OpenCV, JavaScript DCNN face detector	Accuracy 90% on RESNET50	In case of harsh and wild environments, the system does not provide high accuracy

prove to be highly effective [85]. Multiple proctoring systems are actively using cameras to record the activities of the examinees, enabling invigilators to spot any fraudulent actions. Furthermore, artificial intelligence, alongside automation, is revolutionizing the monitoring of human activity during exams. Machine learning algorithms such as Ada-boost [86], Naïve Bayes [87], Random Forest [88], and Bayesian Networks [89] have been employed to detect any malicious activity, and deep learning systems involve image and video recognition, image analysis, classification, and recommendation using Convolutional Neural Networks (CNNs).

Activity recognition remains a research area focused on identifying, tracking, and locating key points such as elbows, eyes, knees, or noses in the case of humans [90]. Unusual behaviors are captured and utilized to identify and prevent potential suspicious activities. This chapter offers a concise overview of various research efforts pertaining to human activity recognition in proctoring systems. Yulita et al. [91] proposed a human activity recognition system based on AdaBoost and SVM. They utilized fixed sampling for data preprocessing and a Butterworth filter to remove the noise. The system employed a dataset of 10,299 images. It compared human activity with algorithms such as Naïve Bayes, Random Forest, Bayesian Networks, Bagging Extra Tree, K-Nearest Neighbor, and Random Trees, achieving an accuracy of 96.06%. However, there is still a need to improve the accuracy further.

Dileep et al. [92] proposed human activity recognition to monitor the suspicious behavior of the examinees. Their system utilized 2D pose estimation, a CNN model, and a custom dataset to detect malicious activity. They employed CNN for classification and a SoftMax function to classify the activity as OpenPose [93] and MediaPipe [94]. However, their work lags in terms of evaluation and implementation details.

Amruthet et al. [95] developed a deep learning-based system for recognizing suspicious activities in surveillance videos. Their main objective was to create a system that automatically detects unusual or abnormal activities. They employed the DeepCNN technique and the KTH and CAVIAR datasets for their analysis. The system was implemented in the OpenCV Python library using a pre-trained CNN model, the VGG-16 neural network. Additionally, they trained their model using the LSTM architecture and achieved 76% accuracy within the first 10 epochs across three classes. However, there is a need to improve the accuracy further.

Genemo et al. [96] presented a system for recognizing suspicious activities of students during examinations. They utilized a 63-layer DeepCNN model called “L4-BranchedActionNet.” The system involved feature extraction using PCA, and classification was performed using SVM and KNN, while feature subset selection was conducted using ant colony optimization. The proposed system was implemented in MATLAB using the CIFAR-100 and CUI-EXAM datasets, achieving an accuracy of 92.99%. The authors suggested further exploration of multiple feature selection techniques and the adoption of feature fusion instead of feature selection methods to improve accuracy and achieve higher performance. Saba et al. [97] introduced a method to recognize suspicious activities in intelligent visual systems. Their approach involves using L4-BranchedActionNet along with entropy-encoded ACS, PCA for feature selection, and SVM and KNN for classification. They implemented this methodology in MATLAB using various datasets such as CIFAR-100, the Weizmann Action dataset, and activity datasets, including HDBM51 [98] and AIDER [99]. The method achieved a high accuracy of 99.24%, improving performance as the number of features increased. However, a large feature set led to increased computation time for the algorithm, posing a significant limitation.

Khan et al. [100] developed a Human Suspicious Activity Recognition (HSAR) system using surveillance videos. The study utilized the CNN model and an optimal features-based framework, which employs deep learning models such as DarkNet 19 and NasNet Mobile through transfer learning. The system also utilized serial-based feature fusion and feature selection via equilibrium feature optimizers and neural network classifiers. Their technique was tested on Hybrid KTH01 and KTH02 datasets and achieved an accuracy of 99.8% and 99.7%, respectively. It was noted that fusion-based features provided highly accurate classifications. However, their proposed methodology overlooked some features, which led to misleading results. Additionally, their model must be tested in real-time to evaluate its efficacy.

Li et al. [101] proposed a suspicious human activity recognition system based on combined 2D and 3D feature selection with LSTM-based networks. Their system aimed to capture complementary information from still and motion frames. The methodology was tested using the UCF101 and HDBM51 datasets and achieved an accuracy of 94.2%, specifically due to the attention-based feature fusion technique. Ni et al. [102] presented a system for analyzing human activity that uses the pose adaptive motion feature pooling technique. Their methodology tackles the spatial and temporal motion issues in videos. It emphasizes using videos and human pose types where the problem has been formulated as a latent structural learning problem. They employ feature selection using Histogram of Gradients (HoG) and SVM for classification. The system has been tested using the HDBM51, KTH, and Hollywood2 movie action datasets, RGBD-HuDaAct Benchmark datasets, and fine-grained kitchen action datasets.

Avola et al. [103] developed a 2D skeleton-based activity recognition system for behavioral analysis. Their approach involved using two-branch stacked LSTM-RNNs, and generating lightweight features from 2D skeletons by means of C-RNN and GANs. The system addressed issues such as partial body occlusion management and multiple subject identifications within a 3D DenseNet framework. They tested their methodology using the KTH, Weizmann, UCF, Kinetics400, HDBM51, and IXMAS datasets, and data augmentation was performed using generator and discriminator frameworks. The

methodology obtained an accuracy of 95.12%, but there is room for improvement to achieve higher performance.

3.3.8 Summary and insights

Activity recognition relies heavily on the feature selection technique employed below. Machine learning algorithms provide richness in accuracy and feature selection in human activity recognition systems. Adaboost, SVM, and Random Forest demonstrate the highest classification accuracy compared to other machine learning techniques. OpenPose and MediaPipe are the pose estimation models that have been actively utilized for activity recognition in various works.

In order to alleviate the limitations of the feature selection technique, feature fusion, and multi-label feature selection-based methods yield high accuracy [104]. Various pre-trained deep neural networks, such as NasNet [100], EfficientNet [105], EfficientDet [106], and other object recognition models, can be employed for activity recognition. Additionally, 3D skeleton-based feature selection methods facilitate efficient motion detection in video frames, which can be utilized for surveillance [107]. Accurate analysis of activity recognition in real-time scenarios can be facilitated by GANs, transformers, and generator-discriminator frameworks. As a result, high-end infrastructure is necessary for real-time implementation of human activity recognition. Table 6 provides a comparative analysis of activity recognition in online proctoring systems, while Fig. 5 illustrates the generic framework for activity recognition in proctoring applications.

3.3.9 User verification

User verification is undoubtedly a critical aspect of automated proctoring. Visual verification, particularly through face recognition, is extremely important throughout an examination session. Continuous user verification involves iteratively confirming user interaction with the system to verify that the person using the system is the same as the one who initially logged in. This process is essential for granting access to computational resources and remote services. The remote service needs to be activated for user verification throughout the session, which can be challenging in distance learning. User verification is highly complex as well as it requires non-intrusive authentication during online tasks and the precision and accuracy of biometric techniques [108]. This chapter highlights some of the prominent works carried out by researchers on automated proctoring systems.

Purohit et al. [109] have proposed an efficient multi-modal biometric fusion user authentication system for remote proctoring applications. Their methodology involves using different soft and spectral-domain biometric features such as mouse, touch, and keystroke from real-time datasets. They employed a Modified Wolf Optimization (MWO) algorithm to compute the matching score values with the assistance of different soft biometric weights. Additionally, they utilized an Optimal Feature Fusion (OFF) algorithm to combine optimal weight features of multiple biometric responses. Moreover, they applied a hybrid LCNN-Salp swarm optimization-based classifier to continuously authenticate the identity of the examinee and monitor their presence and interaction. The modified wolf optimization algorithm is designed using two other optimization methods: the weil optimization algorithm and the gray wolf optimization algorithm. Their method has been implemented using the ORL database, Yale face database,

Table 6 Comprehensive analysis of activity recognition on online proctoring systems

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Yulita et al. [91]	Human Activity Recognition System	AdaBoost, SVM	To Detect Malicious Activity	-	Accuracy 96.06%	Still, accuracy needs to be improved
Amrutha et al. [95]	Deep learning based Activity recognition	CNN, VGG16, LSTM	To detect suspicious activity on surveillance videos	OpenCV, Python	Accuracy 76%	The model identifies the activity however the precision and accuracy is still question mark.
Genemo et al. [96]	Suspicious Activity Recognition System	L4-BranchedActionNet, PCA, SVM and KNN, 63 Layers of DeepCNN	To detect suspicious activity by the examines in examination halls	MATLAB, CIFAR-100 and CUI-EXAM	Accuracy 92.99%	The proposed method does not support multiple feature selection. Feature fusion methods can be adopted in place of feature selection, which can be explored further.
Saba et al. [97]	Intelligent Visual Surveillance systems	L4-BranchedActionNet, Ant Colony Optimization, CNN, SVM, KNN, Entropy	To distinguish the suspicious activity	CIFAR-100, Weizmann action dataset, Softmax function, MATLAB 2020a, Activity datasets HMDB51 and AIDER datasets	Accuracy 99.24% Cubic SVM achieves high accuracy with the activity dataset	An increase in the feature size increases the computation time
Khan et al. [100]	Video based Suspicious Human Activity Recognition system	CNN, DarkNet, NasNet Mobile, Equilibrium feature optimizer	To distinguish the suspicious activity	Hybrid KTH01 and KTH02 datasets, MATLAB	Accuracy 99.8%. Feature fusion based methods achieved high classification accuracy	The model has to be tested for real-time simulations
Li et al. [101]	Human Action Recognition System	LSTM, C3D feature extraction, Spatio-Temporal Feature Selection, Discriminative Part Selection	To capture complementary information from still and motion frames	UCF101 and HDBM51 datasets	Accuracy 94.2% Attention-based feature selection yields high accuracy	Only temporal information is considered, and structural information is omitted
Avola et al. [103]	Video-based action recognition	LSTM, RNN, 2D skeletons, 3D-CNN, GAN	To perform behavioral analysis in action recognition systems	KTH, Weizmann action dataset	Accuracy 95.12%	The methodology is complex and requires high-end infrastructures

FASSEG, KEYSTROKE, and Fingerprint datasets and has achieved an accuracy of 99.12%.

Yusuf et al. [110] developed an automated exam proctoring system that focuses on continuous user authentication through face augmentation and verification. The system's primary objective is to detect cheating behavior and uphold the integrity of

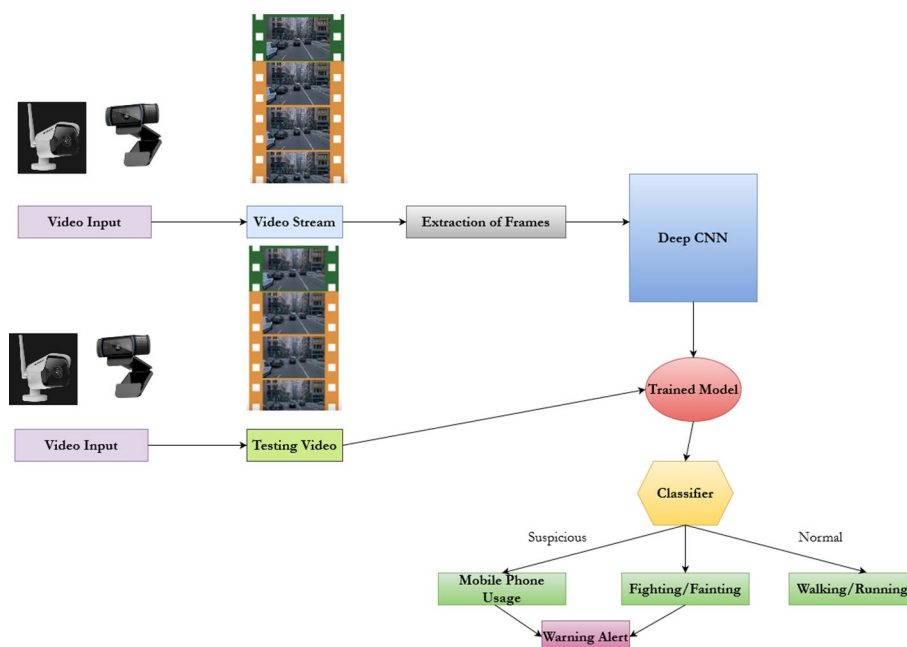


Fig. 5 Generic framework for activity recognition in proctoring applications

automated exam proctoring systems. Their methodology involves utilizing a 3-layer cascaded convolutional neural network for face detection, followed by a face augmentation process that includes generic, component, and attribute transformations. Face verification is then conducted using CNN architectures to enhance user authentication accuracy. However, this process significantly consumes computational time and resources, which degrades the performance of the system. Furthermore, the details of the proposed implementation methodology are not fully disclosed, which raises questions about the accuracy of the system.

Fenu et al. [111] proposed a multi-biometric system designed for continuous student authentication on e-learning platforms. The methodology employs a device and interaction-agnostic approach to continuously verify the student's presence and interaction without disrupting their activities. The system combines biometric data from various sources and uses a score-level fusion method to authenticate the user based on the device used and the interaction carried out with it. While the system is considered reliable and efficient, there is a need for more information on its implementation details as well as real-time testing.

Subramanian et al. [112] proposed a method for continuous verification of test-taker presence during online exams to address integrity issues associated with online exams and reduce test-taker impersonation. They utilized Aadhar biometric data to verify the test-taker's presence throughout the examination session. Their approach eliminated the need for a real-time proctor. However, their system lags in terms of implementation and experimentation details.

Hadian et al. [113] introduced a continuous user verification biometric system for online exam proctoring in M-Learning. The main objective of their system is to reduce and detect cheating. It consists of biometric recognition, where the images are trained with CNN, and a proxy is present in front of the cameras. The system prototype has been implemented with incremental training, followed by an Android application, and

has been implemented via OpenCV, Python, and TensorFlow frameworks. Their system increases resistance to face verification with variations of poses and under different lighting conditions. However, their work lacks significant performance analysis. Sultana et al. [114] presented an innovative framework for detecting malpractice in online proctoring systems. Their method involves identifying suspicious head and mouse movements, categorized as rapid, fast, and trustworthy. They employed face recognition to calculate head movement angles and sides based on features from the Histogram of Gradients (HoG) implemented using OpenCV and local binary pattern histograms in the dlib and MTCNN models. While their method efficiently detects mouse and head movements, it lacks details for comparative analysis.

3.3.10 Summary and insights

Most of the existing methods struggle with implementing and analyzing the performance of user verification systems. Continuous authentication systems that rely on the fusion of multiple biometric modalities require the use of automation, AI-based methods, and optimization techniques in order to analyze user activity [115]. It is evident that continuous user verification encompasses facial, keystroke, fingerprint, and voice biometrics [116]. Hierarchical Multilevel Architectures enable context-aware security by securely handling user verification and session management [117]. These external systems operate based on the user's activity in front of the camera, providing heartbeat-based continuous user verification/authentication [118]. Continuous user authentication based on heartbeat recognition with IoT-based wearable sensors can be explored due to its ease of implementation [119]. Table 7 summarizes and provides insights into the continuous user verification feature in proctoring systems, while Fig. 6 presents the generic framework for user verification in automated proctoring systems.

3.3.11 Authentication and security

To ensure the exam's integrity, authentication in proctoring entails confirming the identity of candidates for online exams. Examinees are verified and validated through various techniques involving distance proctor monitoring [120], identity card verification and validation, and biometric technologies. These techniques seek to verify the test taker's identity before and during the exam [121]. To perform the authentication, various methods such as attribute encryption, bilinear pairing, elliptic curves, symmetric key encryption, identity-based encryption, facial recognition, and biometrics, including heartbeat, palm print, fingerprint, and eye print, are employed. Deep learning methods like CNN and pre-trained deep neural network models are also essential [122]. Additionally, candidates must validate their identity information, display their identity card to a remote proctor for verification, and log in using valid credentials in order to take the exam. This section highlights some of the prominent works in authentication and security in proctoring systems.

Yaqub et al. [123] proposed a privacy-preserving anomaly detection system for video-based, crowd-sourced online proctoring applications. Their methodology involved using an image-hashing method to detect the face and body movements of students engaging in cheating or misbehavior during exams. Their system is designed to identify a student's face, even if it is blurred or masked in video frames. Student verification is performed using the OCR technique, and any videos with obscured faces are sent to the

Table 7 Comprehensive Analysis of Continuous User Verification in Automated Proctoring Systems

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Purohit et al. [109]	Remote proctoring Application	Hybrid LCNN-Salp Swarm Optimization, Modified Wolf Optimization	To ensure multi-modal biometric fusion-based live tracking for proctoring applications	ORL database, Yale face database, FASSEG, KEYSTROKE, and Fingerprint datasets	Accuracy 99.12%	Lags details about implementation details To be tested for real-time implementations
Yusuf et al. [110]	Continuous user verification system	3-layer Cascaded CNN	To detect the possibility of cheating and to maintain the integrity of the examination	Face Dataset	Increases the verification efficiency	Face verification consumes a lot of computational resources and time Additionally, it lags the details of the implementation
Fenu et al. [111]	E-learning Platforms	Score level fusion scheme	To verify the examinee	-	Reliable and Efficient	Tested for real-time implementations
Subramanian et al. [112]	Online Examination Systems (OLEs)	Aadhar	To perform user integrity	-	Eliminate the need for the proctor	Real-time testing is required
Hadian et al. [113]	M-Learning Platform	Biometrics	To reduce and detect the possibility of cheating	-	Increases the resistance of face verification with variation of poses and under different lighting conditions	Performance evaluation is missing in their proposed work
Sultana et al. [114]	Online proctoring systems	Face detection, Recognition and Pose Estimation, Dlib and MTCNN	To reduce the risk of cheating in online exams	-	MTCNN achieves good accuracy	Lags in comparative analysis with the benchmarks

third-party proctor, who then utilizes a tool to detect student anomalies. The students who are involved in cheating will be flagged, and a third party will anonymously review their details before being reported to the university staff. The student used both in-house and publicly available datasets such as HELEN, UTKFace, CelebA, RF, LFW, etc. Dlib and MediaPipe models, along with an image hashing method, were utilized, resulting in 94.4% accuracy. The model demonstrated low computation overhead, but it suffered

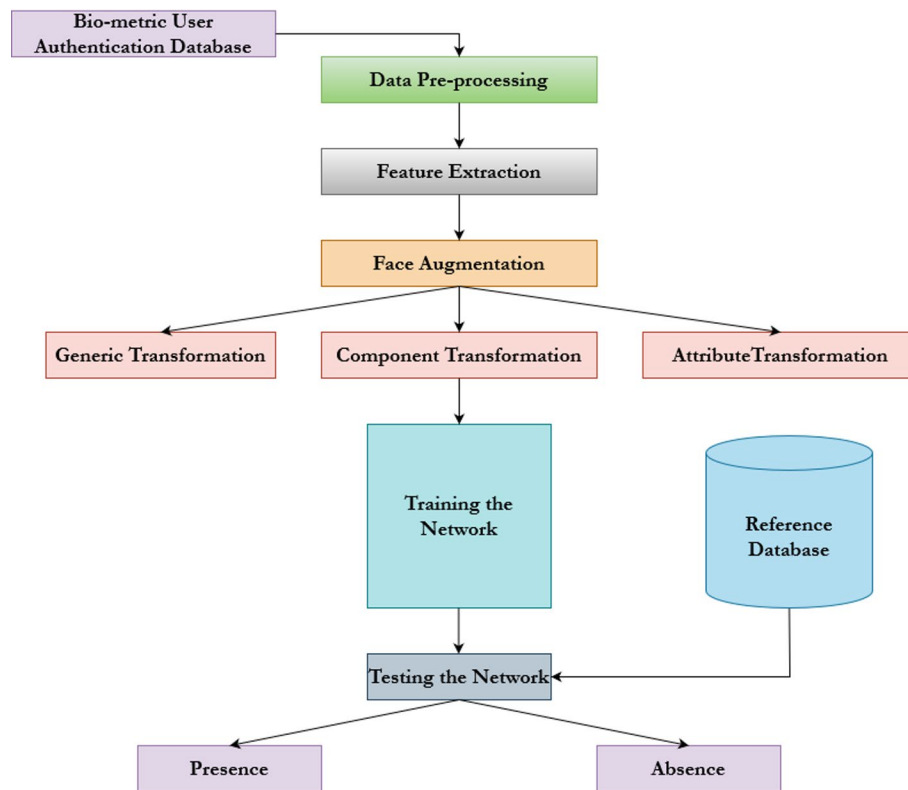


Fig. 6 Generic framework for user verification in online proctoring systems

from long processing times, which could potentially be improved with cropping and simple scaling approaches.

Yaghi et al. [124] proposed a student authentication and proctoring system using AI and IoT technology. In their system, a computing unit equipped with an HD camera and implementing computer vision pipelines is used for identity verification through biometric scanning or identity card scanning, followed by face detection and pose estimation. They employ the Haar Cascade algorithm along with CNN and LSTM classifiers for face detection, and faces are detected using OCR. The system achieved a 70% performance rate, but it's not sufficient for conducive proctoring environments.

Zhu et al. [125] have developed a secure biometric authentication and a blockchain-based framework for online examinations. Their proposed encryption-based scheme aims to ensure the security of biometric features and provide fine-grained access control. The system involves extracting face features using random points and creating a fuzzy vault, and the cipher data is then protected using attribute-based encryption, with access policies owned by the data owners. The encrypted data is stored in a distributed data storage system, and only individuals who meet the access control policies can access the data. Multiple authorities distribute the attribute keys in SEBB. The proposed scheme utilizes bilinear pairing cryptography, where the data is stored and secured using the feature polynomial and majority voting schemes. This approach results in improved communication and computation performance, along with high security, privacy preservation, fine-grained access control, and alleviating central tendency and dispute resolution. However, the scheme does not incorporate physiological biometrics for continuous user authentication.

Amkamaran et al. [126] proposed a secure continuous user authentication scheme to prevent cheating in e-examinations. Their methodology addressed the problem of cheating in online exams. Their system utilizes an AI-based approach and provides a framework for user authentication and security. Their system provides a strategy for identifying and deterring students from cheating during examinations through face recognition using their e-Exam software. This software efficiently captures the screenshot of the student during the examination and reserves it in the database for careful investigation by the proctor. However, the system's performance and accuracy need to be improved. Kulkarni et al. [127] introduced a novel approach for securely conducting online examinations using smart contracts and blockchain technology. Their system leverages smart contracts and Hyperledger fabric. The proposed method incorporates a smart contract and face detection algorithm, implemented with the assistance of the GoCV package for the OpenCV computer vision library. The use of smart contracts ensures high system security. However, their work lacks the implementation details necessary for performance analysis.

Ketab et al. [128] presented a biometrics-based authentication framework designed to provide ongoing user authentication for online assessments. Their system includes a 2D and 3D feature extraction controller, an authentication engine, a biometric profile controller, and an authentication and communication controller that uses a camera and eye tracker. They analyzed the security of the authentication using nine different threat models. However, it is necessary to test their proposed system for real-time implementation. Kabier et al. [129] proposed a role-based access control biometric authentication scheme for online examination systems. Their system involves the use of asymmetric scalar product-preserving encryption and a biometrics-based authentication approach to ensure privacy and security during online examinations in a cloud environment. Their system is designed to resist replay, MitM, and man-in-the-middle attacks while achieving mutual authentication, user anonymity, and forward secrecy. The proposed system has been tested using the Scyther tool with a GUI but still needs to be tested for real-time implementations.

Alnasser et al. [130] have come up with a four-stage security model for conducting e-exams in educational institutions. Their system employs biometrics as the first level of defense, followed by encryption, Anti-X, and SQL injections to secure the data on the cloud. The model has been implemented using the Infrastructure as a Service (IaaS) model to enhance security. It is distributed and provides protection against internal attacks while preventing access to malicious programs and securing isolated end-user machines. However, certain practical limitations exist, such as resource allocation, timely technological updates, and third-party vendors.

Nugroho et al. [131] studied academic integrity and impersonation attacks in online examinations and how to detect them. Their system focused on the reliability and accuracy of face verification. They used novel methods based on similarity scores to verify the validity of the student by comparing their face with the profile picture from LMS. Their system employed facial detection and recognition algorithms, along with machine learning techniques, to ensure security and integrity. The cheating detection pipeline divides the input into images to easily identify static photo impersonation. Their methodology provides a robust solution for detecting impersonation attacks, thereby ensuring the fairness of exams. However, their methodology is dynamic and is not suitable

for live proctoring. Ullah et al. [132] proposed a multi-factor authentication technique to ensure the security and integrity of online examinations. Their study investigated the use of dynamic profile questions in a proctored examination. Their system involves using an ANOVA test to measure its performance. The use of dynamic profile questions decreases impersonation attacks, thereby achieving secure authentication. However, the proposed work needs to be tested with a larger sample size.

3.3.12 Summary and insights

The privacy of students is a significant concern [133, 134]. Currently, no commercially available dataset exists for exam-taking and cheating attempt videos [135]. However, a publicly available dataset has helped identify privacy issues related to anonymized faces. Proctoring can bring forth the threat of bias and discrimination when the proctors make intuition-based judgments based on the behavior or appearance of the examinee during the exam [136]. Online examinations are highly susceptible to impersonation, man-in-the-middle, replay, and other security threats. Sometimes, these systems may also encounter network connectivity issues, which can disrupt the continuation of the exam and affect its output. Methods such as web browser-based authentication, biometrics, physiological-based authentication, sensor-based authentication, and AI-based authentication are also vulnerable to security and privacy attacks. Therefore, it's crucial to explore appropriate encryption that supports object-oriented paradigms and collaborative frameworks [137]. Table 8 provides a comprehensive analysis of authentication and security solutions pertaining to automated proctoring systems.

3.4 Comparative analysis of datasets, features, trends, and technologies of online proctoring systems

The majority of the existing systems are based on deep learning, optimization techniques, machine vision, and other advanced methodologies. Proctoring systems can include eye gaze detection, facial recognition, head pose estimation, activity recognition, user verification, authentication, and security. Each type of proctoring system uses different datasets and involves various trends, technologies, and proctoring tools. This section provides a comparative analysis of various datasets, features, and types of online proctoring solutions. Table 9 provides an overview of the datasets, features, trends, and technologies of online proctoring systems.

4 Comprehensive analysis of security and performance parameters utilized for proctoring systems

This section provides a comprehensive analysis of the security, requirements, threats, and security measures used in online proctoring systems, followed by an evaluation of the performance parameters for proctoring systems.

4.1 Security

Security ensures that data is free from breaches and attacks and guarantees Confidentiality, Availability, and Integrity (CIA). Online proctoring systems face various challenges, such as fraudulent activities, violations of examination rules, disturbances in proctoring operations, embezzlement of personal information, storage disruptions, and compromising the trustworthiness of assessment activities and results. To ensure security,

Table 8 Comprehensive analysis of authentication and security systems for proctoring

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Yaqub et al. [123]	Video-based crowd, crowd-sourced online proctoring systems	OCR, Face Recognition and Pose estimation	To detect the face and body movements of students who are engaged in cheating/mis-behavior during the exam	OpenCV	Achieved an accuracy of 70%	The system is not sufficient to provide security
Yaghi et al. [124]	AI & IoT powered Proctoring System	Computer Vision Pipeline, Dlib, MTCNN, Haar cascade, LSTM, OCR	To perform cheating in online exams	Raspberry Pi	Monitors detect, and reports academic malpractice	Improvement in terms of accuracy needed to be ensured
Zhu et al. [125]	Online Examination Systems	Blockchain, Bilinear Pairing, Random points, Attribute-based encryption	To identify and to detect the issue of authentication	NS3	Achieves better communication and communication costs, privacy, and security	Does not include physiological biometric for continuous user authentication
Amakamran et al. [126]	E-examination system	AI	To prevent cheating in e-Examinations	-	Provides continuous user authentication and security	Accuracy and performance need to be improved.
Kulkarni et al. [127]	Online Examination System	Blockchain, Hyperledger Fabric, Smart Contract, Face detection algorithm	To conduct online examinations in a secure manner	GoCV, OpenCV, Python	Highly secure due to the use of Smart Contracts	Implementation details are not specified
Ketab et al. [128]	E-assessment Applications	Biometrics based Feature Extraction Controller, Biometric Profile Controller, Authentication, Communication Controller, 2D, 3D face recognition algorithm	To identify the potentiality of cheating	Eye Tribe Eye Tracker	Nine threat scenarios have been evaluated	Lags experimentation and performance analysis
Kabier et al. [129]	Online examination systems	Role-based access control, Asymmetric Scalar Product Preserving Encryption (APSE), Fingerprint biometrics	To perform user authentication and end-user monitoring	Scyther tool	Secure against MitM, replay, and insider attacks Achieves mutual authentication, user anonymity, forward secrecy	To be tested for real-time implementations

Table 8 (continued)

Reference	Application	Algorithm	Purpose	Implementation	Benefits	Limitations
Alnasser et al. [130]	E-exam Systems	Biometrics, Encryption, Anti-X, SQL injection, IaaS cloud	To provide security against impersonation, tampering, and data capture attacks	-	The security system is distributed and protects against internal attacks, stops intruders and secure isolated end-user machines.	Resource Allocation, Timely technological updates, and third-party vendors pose practical limitations
Nugroho et al. [131]	Online examinations	Similarity Score, Face Recognition	To achieve academic integrity and to mitigate impersonation attacks	Cheating detection pipeline API	Secure against impersonation attacks	Model is static and not suited for live proctoring
Ullah et al. [132]	Online Examinations	Multi-factor authentication	To mitigate impersonation and abetting in online examinations	ANOVA Test	The proposed system decreases impersonation attacks with live proctoring	The proposed work has to be tested with a larger sample size

appropriate security measures, policies, and rules encompassing physical, technological, and administrative levels of control have to be implemented [167].

4.2 Security attackers

Security attackers can typically be segregated into inside/outside attackers, active/passive attackers, and malicious/rational attackers [168].

4.3 Security assaults

Security attacks in online proctoring systems can be broadly classified into different types, namely masquerading, stealth, malicious (viruses, worms, trojan horses), spoofing, Denial-of-Service (DoS), man-in-the-middle, impersonation, replay, and storage attacks [169, 170]. Data leaks and access control breaches pose risks to confidentiality [171]. Attacks on integrity are related to plagiarism and fraudulent activities [172]. Attacks relating to availability include denial-of-service and deletion of exam logs and recordings, which account for assaults on storage [173]. Impersonation-based attacks involve replacing authorized personnel with a fake person to take the exam [174]. Additionally, man-in-the-middle attacks can be performed during the exam event by recording the exam and breaching the access controls. Lastly, an adversary can disrupt the network once access rights are obtained to break the protocol. Strong control measures should be maintained at various levels to ensure security in online proctoring systems.

4.4 Security requirements

Earlier online proctoring required authentication mechanisms related to single-face, single-user biometric authentication. However, due to scalability issues, it is important to have multi-factor authentication protocols that can support biometrics, face, speech, sound, and iris recognition on a larger scale [175]. Similarly, adherence to CIA principles

Table 9 Comparative analysis of datasets, features, trends, and technologies in online proctoring systems

Type of solution	Reference	Datasets used	Trends and technologies	Features
Eye Gaze Estimation	Liu et al. [31]	CUI Exam Dataset, Self-generated Video, Image Dataset	Camera	Action Recognition, Cheating Behavior Detection
	Yaqub et al. [32]	Self-generated Exam-taking Video Dataset [138]	Camera, OCR, Eye Calibration	Text Detection and Verification, Anomaly Detection, Face Recognition, Gaze Estimation
	Dilini et al. [35]	Self-generated Eye Gaze Dataset	Webcam, Browser, API, WebGazer	Cheating Behavior Detection, Gaze Model, Browser Locking, Eye Tracking,
	Ramachandra et al. [38]	Self-generated Eye Gaze Dataset	MCQs, SMI Redn Professional [139] eye tracker, PyGaze [140]	In-attentional Blindness, Wavering Behavior,
Head Pose Estimation	Potluri et al. [55]	Crime Investigation and Prevention Lab (CIPL) Dataset, CASIA WebFace Datasets	Video, Camera, Head Poses	Face Detection, Face Alignment, Face Spoofing, Mobile Detection, Multiple Person Detection
	Chuang et al. [56]	Self-generated Video Dataset	Visual Focus of Attention, Head Poses, Camera	Cheating Behavior Detection,
	Liu et al. [59]	Pointing '04 [141], AFLW [142], HRIHP Datasets	Camera, Head Poses,	Facial Recognition
	Zhang et al. [61]	300 W-LP Dataset [143], AFLW Dataset [144], BIWI Dataset [145]	Camera, Images, Head Poses	Face Detection
Facial Recognition	Liu et al. [62]	AFLW Dataset, BIWI Dataset, 300 W-LP Dataset	Camera, RGB images, Head Poses, MTCNN Face Detector	Facial Feature Identification, Human Behavior Analysis
	Liu et al. [64]	IRHPE, CAS-PEAL-R1 Datasets, Pointing'04 Datasets	Camera, Head Poses, angles	Face Texture Extraction
	Liu et al. [65]	CAS-PEAL-R1 Datasets [146], Pointing'04 Datasets	Camera, Head Poses,	Face, Human-Computer Interaction
	Elharrouss et al. [70]	LFW [147], CFP [148] and CASIA-WEBFACE [149], pointing'04 and Schneiderman [150]	Camera, Head Poses	Image Recognition, Face Recognition
	Ganidisasthra et al. [72]	Self-Generated-Facial dataset	Camera, Face angles, Face Poses, Images	Face Detection, Face Estimation, Face Verification
	Ozdamli et al. [75]	LFW [151], FER [152, 154], Gl4E [153], FEI Face Datasets and COCO dataset	Sensors, Cameras, Images, Emotion Capture	
	Adhikary et al. [79]	MPH [155], CKPlus [156], Facial Keypoints Detection Task [157] Datasets	End-to-End Pipeline, Webcam, Poses, Images	Emotion Detection, Cheating Detection, Face Recognition, Sentimental learning, Human Behavior Analysis, Face Keypoint Extraction
	Yang et al. [81]	Self-generated Dataset	OCR, Camera, Webcam, Videos, WebAPI	Exam Monitoring, Analysis, Face Detection and Recognition, Face Classification

Table 9 (continued)

Type of solution	Reference	Datasets used	Trends and technologies	Features
Activity Recognition	Genemo et al. [96]	CUI-EXAM [158], CIFAR 100 Datasets [159]	Webcam, Videos, Data Augmentation	Cheating Behavior Detection, Back watching, Front Watching, Showing Gesture, Side watching, Suspicion
	Saba et al. [97]	CIFAR 100 Datasets, HMDB51, Weizmann and AIDER Datasets	Webcam, Video, Images	Suspicious Behavior, Cheating Detection, Human Recognition
	Khan et al. [100]	Public [160], CAVIAR [161], Hybrid-KTH01 and Hybrid-KTH02 Datasets	Webcam, Video, Images	Cheating Detection
	Li et al. [101]	UCF101 [162], HDBM51 datasets	Camera, Images	Human Behavior Analysis, Cheating Detection
	Avola et al. [103]	KTH, Weizmann Datasets, UCF, sports, IXMAS, HDBM51, UT-Kinect, UCF101 and Kinetics 300 and NTU RGB + D datasets	Camera, RGB Images, Video, 3D Skeletons	Multiple Subject Identification, Face Recognition
User Verification	Purohit et al. [109]	ORL Database [163], Yale face database [164], FASSEG [165], KEY-STROKES [166], finger print datasets	Keystrokes, Iris, Speech, Face, Webcam, Biometrics, Authentication, RGB Images, Optical Sensor	Face Recognition, Iris Recognition, Biometrics Identification, Keystrokes Dynamics, Speech Recognition, Emotion Recognition
	Yusuf et al. [110]	-	Face Images	Face Augmentation, Face Recognition, Exam Proctoring, User Authentication, Cheating Detection
	Fenu et al. [111]	Biometrics Iris Dataset	Biometrics, Face, Voice, Mouse, Touch, Keystroke, Student Identity, Keystrokes	Student Authentication, e-learning, Mobile App, MCQs, Cheating Detection
	Subramanian et al. [112]	Aadhar Data	Multi-Biometrics, Cloud, LMS,	e-Assessment, Cheating Detection, Proctoring, User Authentication
	Kulkarni et al. [127]	-	Smart Contract, Auto-Grading, Web Application, Desktop Application, Blockchain, Face, Mouth	Online Examination, MOOCs, Authentication
Authentication and Security	Ketab et al. [128]	Self-generated dataset	2D Facial, 3D Facial, Biometrics, Camera, Mouse, Keyboard, Mic, Eye Tracker, Speech, Eye, Head Movements, Eye Tracker, Camera, Videos, Eye Tracker, Sound, Sensor	E-assessment, 3D Face Recognition, e-learning, Electronic Invigilation, Speech Recognition
	Alnasser et al. [130]	-	Key Strokes, Cloud, IaaS	Cheating Behavior, e-Exam3
	Nugroho et al. [131]	-	Virtual Webcam, Static Photos, Seminars, LMS, Videos	Face Recognition, Pipelines, Attack Detection

and privacy is also crucial to ensure academic integrity [176]. Safeguarding the original identities of test-takers is imperative, and institutions conducting the exams need to uphold privacy. Additionally, anonymity should be maintained at the administration level, access level, and other levels. Biometrics must be protected to ensure identity

privacy [177]. Appropriate security techniques, including cryptographic techniques, are needed to protect the system and privacy [178]. Moreover, enhancing computation and communication costs will improve the efficiency and performance of the system.

4.5 Security controls

Security controls enable us to achieve the integrity and dignity of the teaching and learning experience. In order to ensure security, typical controls inside the software are highly essential for proctoring. Typically, there are three types of security controls, namely administrative, technological, and technical controls, which we will discuss in this section.

4.5.1 Administrative controls

Policies against plagiarism, exam processes, practices, and regulations are examples of administrative controls. Legal regulations can carry heavy penalties for violations. Administrative policies such as the General Data Protection Regulation (GDPR), Service Level Agreements and Policies (SLAs and SLPs), and U.S. privacy shields aim to safeguard personal information and user privacy. By enabling administrative controls, biometric and behavioral data can comply with GDPR regulations.

SLAs and SLPs involve the use of control reports certified by a public accountant to ensure security requirements such as confidentiality, integrity, availability, processing, and auditing. SLAs and SLPs provide control over the system by monitoring activities pertaining to malicious activities, issuing warning alerts in case of aberrations, generating an audit trail, investigating the source of an attack, and implementing countermeasures before any data loss occurs.

The U.S. privacy shield frameworks are employed to design and implement strategies that adhere to data protection requirements and ensure privacy. Automated online proctoring solutions must comply with EU-wide regulations to ensure privacy protection, security, and reliability. The PSI Bridge platform [179] utilizes proprietary browser lockdown and self-authentication to ensure security and privacy compliance and protect against security threats. Cloud-based Infrastructure-as-a-Service (CIaaS) enables the security of the underlying LMS, allowing integration with the learning platform. Online proctoring applications must also incorporate DMCA-based copyright protection.

4.5.2 Physical controls

Physical control involves the ability to detect nearby objects in the test-taking environment and to understand how these objects may pose a threat to confidentiality and academic integrity [180]. Spatial controls with 360° monitoring involve the use of a webcam, screen-sharing capability, and a smartphone camera to provide a complete view to the user [181]. Additionally, facial recognition is used to identify cheating behavior related to unauthorized device access. Video recording, location tracking, audio recording, speech recognition, and noise suppression are essential features.

4.5.3 Technological controls

Controls pertaining to technology guarantee exam integrity and confidentiality. These controls are implemented on data points, computers, networks, software, and physical devices. They can be static or dynamic. Static controls, which are simpler and do

not change, include user biometrics, encryption standards, and browser lockdown. On the other hand, dynamic controls involve the functions and processes necessary for the entire system to operate and include live image capture and activity logs.

In proctoring software, essential controls involve enabling kiosk mode, disabling shortcut keys, right-clicking, copying and pasting, switching to other applications, and browsing. Additionally, features such as flagging, emotion recognition, and speech recognition controls are used to create a controlled environment for proctoring, along with exam video recordings to ensure integrity and confidentiality throughout the session. Data protection is ensured through the use of hash functions, encryption algorithms, digital certificates, signatures, and protocols [182].

4.5.4 Advanced features to prevent fraud

AI-enabled online proctoring systems involve adding new interactive components that identify and detect new behavioral, sentimental, physiological, and physical signals of a user sitting before the screen while taking the test. These movements provoke the proctor's action on the screen or trigger communication through the webcam or the monitor with a student who is involved in cheating or exhibiting abnormal behavior. The use of technological controls can enforce privacy preservation and consent based on SLAs and SLPs. Administrative controls enable location and identity privacy.

- (i) Port Blocking: In order to prevent cheating, all network ports are blocked to control communication and restrict access by using VPN clients, which prevent access to external websites. These controls are administered by the IT managers within the organization. The safest way to run applications separately from the rest of the computer environment is by using Virtual Machines (VMs).
- (ii) Webcams: During a test session, the webcam records the user's activities, which are then used by the proctor to detect the malicious behavior of the student. The video recordings are analyzed for facial recognition, mouth movements, iris recognition, head pose, eye gaze, speech, and audio detection. To expedite this process, an AI-based flagging technique can be employed to identify and flag any abnormal behavior. The webcam also efficiently monitors the space surrounding the test-taker's environment. However, webcam recordings suffer from limitations such as obscure views, low lighting, and occlusion problems.

Face recognition during live proctoring is challenging and highly susceptible to false injection attacks. Implementing webcam-based authentication is also difficult, and controlling the physical environment around the test-taker is not possible.

Any abnormality can be flagged and brought to attention through live, automated proctoring. A student may receive a signal from an undetectable source that falls outside the webcam's range. Access Control Lists (ACLs) in Wireless Access Points (WAPs) help filter devices, limiting the number of connections by their MAC addresses, which would enable the blocking of unwanted endpoints [183]. Visual space controls provide an efficient way to prevent cheating.

- (iii) Browsers: Attacks through web browsers and advertisements may pose a threat to the test taker. Locking down all browsers will efficiently control unauthorized access, preventing test-takers from cheating. Additionally, inducing an alarm system to alert

about any potential cheating behavior will facilitate efficient detection. However, vulnerabilities in the exam environment still need to be addressed.

- (iv) **Facial Emotion Recognition:** The emotions of a student can vary when they are involved in cheating, which can help identify whether the test-taker is cheating or not. This concept was first described by Sown et al. [184]. AI and ML techniques enable a comprehensive understanding of a student's behavior during examinations. Using motion spots in an image sequence can help detect facial expressions. Additionally, static models [185], facial coding [186], facial motion sequences [187], optical flow [188], active contour models [189], probabilistic models [190], adaptive appearance models [191], eigen spaces [192], robust filters [193], facial motion extraction [194] velocity estimation [195], correlation filters [196], and various other methods can be utilized for this purpose.
- (v) **Mobile Detection:** Using multithreading on dlib's facial key points and the distance between the nearby objects, the COCO dataset can be trained using the Yolov3 pre-trained model to detect people and mobile phones [197].
- (vi) **Attentiveness Calculation:** Landmark localization [55] and face spoofing can be used to identify the confidence level that indicates the liveness of a face. First, the frames in the facial image are detected and extracted. Then, features are extracted from the aligned faces. Landmark localization allows the extracted features to identify the liveness and attentiveness of students during exams and in classrooms.
- (vii) **Auto-Grading:** In online proctored exams, auto-grading refers to the automated assessment of students' answers without the need for human interaction. It involves sensing different activities such as keystrokes, mouse movements, and audio/video inputs. Autograding uses AI and NLP [198] but is unsuitable for all types of questions and exams. Excessive facial expressions and body movements are detected with the help of image-hashing techniques [123].

4.6 Performance parameters

The effectiveness of proctoring systems is evaluated by using performance metrics. The efficiency of the proctoring systems is directly proportional to the accuracy and precision of the working methodology underneath. There are different metrics to evaluate the performance aspects of an online proctoring system. This section enables us to understand the performance metrics pertaining to online proctoring systems.

4.6.1 Eye gaze estimation

Proctoring systems use various applications, with a focus on eye movements, to detect cheating during exams. Traditional proctoring involves physically monitoring students to identify cheating. However, in the case of online proctoring systems, through the utilization of web camera recordings, it is possible to flag and warn of potential cheating, either in real-time or when reviewing recorded videos. Many applications pertaining to eye gaze estimation [31, 32, 35, 38, 39, 40] use performance metrics such as accuracy, precision, recall, performance time, training accuracy, training loss, validation accuracy, and validation loss. Table 10 offers a comparative analysis of the performance metrics utilized in proctoring applications for eye gaze estimation.

Table 10 Comprehensive analysis of performance metrics utilized for eye gaze estimation applications

Application	Ref	confu- sion matrix	Elapsed time	Ac- curacy (%)	Preci- sion (%)	Re- call (%)	Training accuracy	Train- ing loss	Validation accuracy	Val- ida- tion loss
Eye Gaze Estimation	[31]						Yes	Yes	Yes	Yes
	[32]			Yes	Yes	Yes				
	[35]	Yes	Yes							
	[40]			Yes	Yes	Yes				

4.6.2 Head pose estimation

Head pose estimation is the process by which a computer analyzes photos or video series to predict the position and posture of the human head in three dimensions. The analysis is relative to the camera, and the head pose changes are typically considered rigid body transformations [65]. First, when maintaining a constant core posture, changes in head pose images are distinct in the yaw and pitch angles. Second, when the fixed angle interval increases, the image fluctuations in the yaw angle direction first grow and subsequently diminish. The position of the head is a crucial clue for assessing human attention, intention, motivation, and more. For example, tracking the driver's head can help estimate the driver's gaze zone, providing an effective method for monitoring focused driving [199]. Moreover, it is commonly known that HPE is widely used in a variety of disciplines, including gesture recognition [200], social communication skill development [201], and pedestrian position prediction [202]. Different performance metrics are used to measure head pose estimation, including head pose angles such as pitch, yaw, roll, mean absolute error, and speed for assessing attention in HCI [62]. Commonly used metrics for head pose estimation include accuracy, recall, F1-score, training loss, precision, and time. While classification rate based on head pose estimation is rarely used [58], mean absolute error is the most common metric [59, 61–63]. Table 11 offers a comparative analysis of the performance metrics utilized for proctoring applications in head pose estimation.

4.6.3 Facial recognition

Face detection and recognition are significant components of the proctoring system. Feature vectors play a major role in face recognition, which involves identifying a face from an image containing multiple faces. Face detection is highly difficult due to the non-static nature of faces, as they vary in size, shape, color, and other features. Therefore, in order to assess the performance of the face recognition application in proctoring, several performance metrics are utilized, including accuracy, recall, precision, and F1-score [72, 81]. Additionally, true positive rate and true negative rate are used to evaluate the algorithm's ability to identify positives and negatives [77]. The confusion matrix is employed to measure the performance of the classification model pertaining to a particular class [74]. Table 12 provides a comprehensive analysis of the performance metrics utilized for proctoring applications in facial recognition systems.

4.6.4 Activity recognition

Human activity recognition is complicated since it involves the semantic classification of motions and gestures. In the online proctoring system, human action recognition aims

Table 11

[illegible]

to automatically empower computers to recognize human actions in real-time videos. In order to detect suspicious activity during proctoring, capturing the spatial and temporal information from still and motion video frames is of high importance. Performance metrics such as accuracy, confusion matrix, true positive rate, false positive rate, sensitivity, specificity, precision, recall, and F1-score are used to assess the performance of human activity recognition. Additionally, AUC curves and ROC curves are utilized to assess the classification performance of the machine learning model [91, 95, 96, 97, 100, 101]. Table 13 presents a comprehensive analysis of the performance metrics employed for proctoring applications in activity recognition systems.

4.6.5 User verification

User authentication or verification involves the appointment of a human proctor to observe the examinee during proctoring. It requires analyzing the video recorded during the exam for any suspicious activity. User verification involves validating the test taker's activity and checking for any impersonation [203]. The most distinguishing metric employed for verification is analyzing the face and head pose angles. To measure the performance of the user verification systems, face biometrics, keystroke dynamics, and speech recognition are used. Metrics such as true positive rate, false positive rate, true negative rate, true positive rate, precision, recall, F-score, accuracy, and specificity are also used. Table 14 provides a comprehensive analysis of the user verification system in proctoring applications.

4.6.6 Authentication and security

Security is a critical aspect of the proctoring system, which heavily relies on the implemented security features. Various technical and technological controls, including firewalls, anti-virus software, browser locking, keystroke dynamics, facial biometrics, fingerprints, facial points, and key points, are utilized to ensure authentication and security. X.501 certificates and Microsoft-based antivirus software help eliminate physical, inside, and outside threats. Additionally, performance metrics such as computation and communication costs are considered. Table 15 provides a comprehensive analysis of security systems in proctoring applications.

5 Commercially available proctoring tools

5.1 Comparative analysis of commercially available proctoring tools

Schools, colleges, and universities have been using commercially available online proctoring tools to conduct remote teaching, learning, and examinations. In the current era of online learning, an automated smart proctoring tool has become the need of the hour [204]. It has been identified that a typical automated smart proctoring system comprises eight important components: a camera, microphone, human proctor, screen sharing and recording, application lock, biometric verification, gaze tracking, and random question banks. An automated smart proctoring system or software can be human-based, fully digital, or a combination of both.

Some of the commercially available open-source proctoring tools include ProctorTrack, Talview Preview, ProctorExam, MapleLMS, ExamRoom AI, Honorlock, Proctorio, Speedexam, Think Exam, and Mettl Examine [205]. These software offer a wide range of features for monitoring examinations even remotely in a live manner [206].

Table 12 Comprehensive analysis of performance metrics utilized for face recognition applications

Application	Ref	Time	Accuracy (%)	Training loss	Detection rate	Prediction (%)	Training time	False positive rate	True positive rate	Recall (%)	Specificity	F1-score	False negative rate	True negative rate	Confusion matrix
Facial Recognition	[70]		Yes												
	[72]	Yes	Yes		Yes		Yes								
	[73]					Yes				Yes					
	[74]		Yes			Yes	Yes	Yes	Yes	Yes		Yes	Yes	Yes	Yes
	[76]		Yes												
	[77]		Yes		Yes	Yes		Yes	Yes	Yes	Yes	Yes			
	[81]	Yes	Yes	Yes	Yes			Yes	Yes	Yes					

This section provides a review of the aforementioned commercially available proctoring tools and software available for online proctoring systems. Table 16 categorizes the software, and Table 17 provides a comprehensive analysis of available automated proctoring software.

5.2 Comparative synthesis of different platforms

With the passage of time, automated proctoring systems have evolved from simple browser lockdown mechanism into sophisticated AI-driven environment. Currently the proctoring industry is led by few key players who have established dominance. A comparative synthesis of the top three platforms— Proctorio, Examity, and Think Exam is depicted in Table 18. Making the right choice of proctoring platform is a crucial task, it must be decided after an in depth analysis of needs, a decision criteria for proctoring platform selection is tabulated in Table 19.

5.3 Conceptual scenarios

The traditional paper-based assessments have undergone significant changes due to the pandemic and technological advancements. Human-based proctoring and learning have transitioned to semi-automated and fully automated online proctoring systems. Currently, semi-automated online proctoring systems are in place, involving both human and online assessments. With the emergence of technologies like blockchain, cloud computing, machine learning, deep learning, and artificial intelligence, proctoring systems have advanced to include human activity recognition, emotion recognition, cheating detection, speech detection, and recognition. Additionally, advancements in big data, data analytics, and statistics have led to the introduction of auto-grading, emotional intelligence tests, audio-visual ability assessments, Intelligence Quotient (IQ) based assessments, and perception-based assessments. The progression is now towards fully automated online proctoring systems, which operate without any human involvement. This section presents conceptual scenarios providing insights into online proctoring systems and their evolution across various paradigms, addressing related issues.

5.3.1 Scenario 5.1 What are the various classes of proctoring assessment tests possible in online proctoring systems?

Since educational institutions are new to online proctoring and learning systems, there has been a lot of resistance in accepting the new emergence by educationists. As a result, students' learning capabilities and beliefs in the curriculum are being impacted. This has led to the mulling up and adoption of inappropriate curriculum choices, resulting in heavy reliance on grading systems. Online proctoring systems are not only challenging for students but also for teachers. This has prompted changes in policies and practices pertaining to online proctoring applications. Instead of paper-based assessment tests, alternative evaluation forms have been structured, as depicted in Fig. 7.

- (1) Assignments: These assessments will be based on logical and rational thinking questions to evaluate the student's grasp of the previous lectures. The assessment questions will test analytical ability and may include speech presentations to gauge vocabulary and evaluation skills. Students will benefit from audio-visual presentations to build confidence in front of a camera. Posting videos on blogs and vlogs will help students develop professionally and improve their expression and editing skills.

Table 13

[illegible]

Group assignments and discussions will be used to enhance creativity and group productivity. Additionally, gadget-based learning has made proctoring and learning more engaging for students.

- (ii) **Self-check Tests:** The questions can be framed by the teacher who taught the course. The test will include logical and analytical reasoning questions, with the main aim of assessing the student's comprehension and the teacher's ability. This test also provides clarity on how thoroughly the students understand the concepts in both theory and practice. Quizzes, polls, and surprise tests can be used to conduct these assessments. Quizzes can include Yes/No questions, multiple-choice questions, one-word answers, or fill-in-the-blanks. Polling can be done using smart boards during or after the subject has been taught.
- (iii) **Group Discussions:** These activities help students understand the concepts thoroughly, providing opportunities for clearing doubts, sharing ideas, socializing, and developing problem-solving skills. The students are segregated into groups, and the teacher assigns a topic based on the subject. Each group presents their understanding, and the teacher can assess and give marks based on the points shared. However, this approach may not be efficient for online learning during the COVID-19 pandemic.
- (iv) **Peer-Tutorial Videos:** Peer-tutorial videos cover conceptual discussions and problem-solving. These videos help students enhance their technological, editing, presentation, and technical capabilities. After recording the videos, students present them to their peers, who then raise queries for assessment. Strict time limits are set for the presentations.
- (v) **Case Studies:** Students are assigned case studies to evaluate their data-handling skills and reasoning capacity. The data is examined using experiments with simulation tools, machinery, or software, enabling students to offer diverse perspectives and a wide range of research ideas.

5.3.2 Scenario 5.2: What is the recent technology that provides efficient proctoring capability for head pose estimation, eye gaze estimation, and activity recognition for online proctoring examinations?

Solution: Recent technological advancements such as ICT and IoT have made it possible to conduct examinations anywhere, at any time, through proctoring. Due to the development of machine learning, artificial intelligence, and deep learning technologies, computer vision has become a prominent arena that provides solutions catering to various needs. Using Computer Vision (CV) in the classroom helps to identify cheating during examinations and maintain academic honesty [216]. Computer vision-based proctoring enables both distant and online proctoring to be more feasible. Computer vision in online proctoring systems involves three different stages: the data acquisition phase, the data processing phase, and the data analysis phase. In the data acquisition phase, either real-time live video is captured, or in the data processing, an image processing algorithm flags the student's malicious behavior automatically with code implemented in the background. In the final data analysis phase, the frame/images are analyzed by a human proctor, and the cheating/non-cheating behavior is established. Recent advancements in deep learning, especially CNNs, have paved the way to verify and validate the identity of students during their online examinations [217, 218]. CNN is tuned and trained by behavioral and face recognition models. This identifies cheating behavior by using the

Table 14 Comprehensive analysis of performance metrics utilized for user verification applications

Application	Ref	Accuracy	Speed	Pose angles	TPR	FNR	FPR	FNR	Precision	Recall	F1-Score	Specificity
User Verification	[109]	Yes			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	[110]	Yes	Yes									
	[114]			Yes								

student's facial expression data. The facial expression involves head movement tracking, eye gaze estimation, and activity recognition via motion detection [219]. A motion detector is coded and executed during video processing, where the frames are sequentially scanned to detect motions. The emotions and sentiments of the students may also be analyzed with the help of emotion recognition models based on sentimental datasets [220]. Several different kinds of cheating modes can be identified depending on the requirement.

An AI-based automated online proctoring system utilizes Viola Jones and RetinaFace [221] CNN for object detection, Lucas Kanade [222] for optical flow, the Kalman filter [223] for real-time object tracking, and the pixel displacement statistical model [225] for cheating detection in a real-time video stream. Facial detection involves Haar Feature selection, Adaboost, and the FaceNet model for face verification. Activity recognition methods include emotion recognition, eye gazing, blinking, closure duration, head movement, and motion detection features. Lucas Kanade is used for head movement and pose estimation by detecting and tracking the septum, eyelids, and mouth outlines [224]. The active appearance model [226, 227] recognizes eye motions and gazes using feature points, while the pixel displacement strategy helps forecast motion detection. Additionally, micro-expressions can reveal human intentions [228], and 3D and 2D head pose estimation models facilitate the recognition of various head poses [229, 230, 231, 232].

Limitations: The amount of storage required for processing proctoring videos poses a major impact on the performance of the system. It can cause delays, breakdowns, malfunctions, and system failures. Real-time video streaming requires personalized, unlimited cloud storage. Training the CNN for facial expression detection, eye gaze estimation, and head pose estimation leads to significant delays, posing a challenge for real-time implementation of autonomous proctoring systems. Blockchain technology offers real-time streaming and distributed data storage, meeting the needs of the future [233].

5.3.3 Scenario 5.3 What technology will be utilized in the future for remote or automated proctoring that involves fine-tuned capabilities for cheating detection during examinations?

Solution: The future driving technologies include virtual reality, augmented reality, mixed reality, and the metaverse. This scenario briefly elaborates on these different technologies, their significance, and their impact on proctoring.

(i) **Virtual Reality:** It provides indispensable information about the user's concentration, awareness, and emotional responses. Comprehending eye movement patterns can assist VR developers in optimizing the placement of information, making sure that important aspects capture users' attention. There are three fundamental categories of eye movements proposed by Duchowski et al. [234]:

Fixations: These occur when the eye lingers over a spot or object to gather visual data. The length of the fixation is often flexible, and more visual information is gathered and processed the longer the fixation.

Saccades: Quick, jerky eye movements that cause a sudden shift in the place of fixation are known as saccadic motions. These movements occur quickly and temporarily impair eyesight. While they are less significant than fixations in eye tracking, they disclose details regarding the user's gaze direction, fixation order, and their visual attention span.

Table 15 Comprehensive analysis of performance metrics utilized for authentication and security applications

Applications	Ref	Accuracy	Precision	Detection rate	Recall	False rejection rate	Communication cost	Computation cost	Average normalized error
Security	[123]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	[125]								
	[128]								

Table 16 Category of online proctoring software

Tools	Live proctoring	Fully automated
ProctorTrack [207, 208]	Yes	No
Talview Preview [209]	Yes	No
Proctor Exam [210]	No	No
Maple LMS [211]	Yes	Yes
ExamRoom AI [212]	No	No
Honor Lock [213]	Yes	No
Proctorio [214]	No	Yes
Speedexam	Yes	No
Think Exam	No	No
Mettl Examinee [215]	No	No

Smooth Pursuits: Slower eye-tracking motions intend to maintain a moving stimulus at the focal point. Data capture and analysis of eye movements must include testing conditions, which are of the following two types:

1. Head-free: These are the setups or systems that allow the user's head to move freely without being restricted to a certain position.
2. Head-still: This condition usually describes an instruction or prerequisite when a subject or participant is expected to maintain their head immobile or stationary.

Limitations: The development of virtual reality has set the stage for the use of eye-tracking technology [235]. Currently, integrating eye-tracking into VR headsets is still not feasible. In proctoring, using VR poses issues such as the need for external hardware [236], as well as accuracy with deviations and errors [237, 238, 239]. The primary objective of VR in proctoring is to measure the attention of the student, and accuracy has to be the factor in determining the student's state of mind. However, a range of gaze patterns are supposed to be analyzed to understand cognitive or behavioral impacts, where the precision and accuracy of eye tracking are highly critical. Motion sickness is an additional issue faced by gaze estimations that leads to elevated drop-out rates due to abrupt head movements or improper adjustments [240]. The use of CNNs is hampered by limited memory and reliance on pre-trained networks.

(ii) **Augmented Reality:** Telestration uses advanced communication techniques, such as voice guidance, visual aids, or video conferences, to deliver guidance and instructions remotely. It involves overlaying hands or other objects on images or videos to facilitate the creation of freehand markups, also known as annotations and instructions [241]. This can be viewed in either 2D or 3D mode with augmented or mixed reality [242]. This technology allows for live streaming of a proctor's hands onto the operating screen, enabling direct/live interaction during examinations. Telestration makes it possible to

Table 17 Comprehensive analysis of commercially available proctoring softwares/tools

Tool	Biometric ID/photo verification	Facial/screen/mic/webcam recognition	Scalability	Auto-mated recording	Browser lockdown	Live recording	Integration (LMS/Board/Moodle/Canvas)	Report & analysis	Human proctored exams	Storage	Privacy & security	24×7 Support
Proctor-Track [207, 208]	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	180days	SSL/TSL 2048 Bit	Yes
Talview Pre-view [209]	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	6 Months	Two Factor Authentication	Yes
Proctor Exam [210]	ID	Screen/Webcam/Mic	5000	Yes	Yes	Yes	Yes	Yes	Yes	-	AES-256/SSL/TLS	Yes
Maple LMS [211]	Yes	Facial-No Yes Screen-No	100k	Yes	Yes	Yes	Yes	Yes	Yes	-	AES-256	Yes
Exam-Room AI [212]	Fido Technology-Yes	Yes	100k	Yes	Yes	Yes	Yes	Yes	Yes	6 months	AES-256	Yes
Honor Lock [213]	Yes	Facial/Screen-Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	12 months	AES-256	Yes
Proctorio [214]	Yes	Facial-Yes Mic, Screen, Webcam-Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	2 or 3 Years	AES-256/GCM & ZKE	Yes
Speed-exam	Yes	Yes	3500 attempts	Yes	Yes	Yes	Yes	Yes	No	6 months	256-bit	Yes
Think Exam	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	1 Month	SSL	Yes

Table 17 (continued)

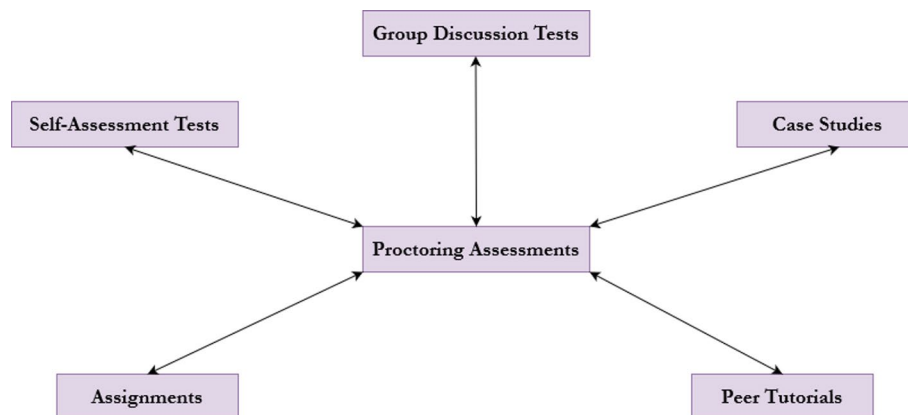
Tool	Biometric ID/photo verification	Facial/screen/mic/webcam recognition	Scalability	Auto-mated recording	Browser lockdown	Live recording	Integration (LMS/Board/Moodle/Canvas)	Report & analysis	Human proc-tored exams	Storage	Privacy & security	24×7 Sup-port
Mettl Exam-inee [215]	Yes	Yes	-	Yes	Yes	Yes	Yes	Yes	Yes	-	256-Bit SSL	Yes

Table 18 Unified cross-platform comparison

Platform	Technology maturity	Scalability	Cost efficiency	Privacy compliance	Use case suitability
Proctorio	High (Fully AI)	Very High	High	GDPR, ZKE	Large-scale academic
Examity	High (Hybrid)	High	Medium	GDPR, Multi-factor	Professional certification
Think Exam	High (AI+Human)	Very High	Medium	SSL, Multi-modal	Government/Corporate

Table 19 Decision criteria for proctoring platform selection

Parameter	Weight	Aspects
Security Requirements	25%	• Identity verification robustness • Data Encryption Standards • Privacy and regulatory compliance
Examination Scale	20%	• User capacity • Geographic distribution • Frequency of testing and examinations
Inclusivity and interference potential (immediate v/s post-exam)	20%	• Requirement for immediate interference • Adjustment for differently abled population groups, ethnicities and cultural background
Technical Infrastructure	15%	• Student device capabilities • Internet bandwidth availability • IT support capacity
Budget	10%	• Per-exam cost vs. subscription models • Infrastructure investment requirements • Training and support costs
Accuracy requirement	10%	• Acceptable false positive/negative rates • Detection sensitivity requirements • Review and appeals process needs

**Fig. 7** Assessment tests pertaining to online proctoring systems

use these teaching methods globally. 3D ghost tools assist with augmented reality, along with robots [243] and 3D avatars [244].

Limitation: Exclusively 5G networks are required since they offer very low latency, which is crucial in various geographical locations. However, the high costs and security infrastructure present significant limitations in implementing this technology.

(iii) **Metaverse:** Metaverse encompasses a broad range of technologies, including 3D modeling, blockchain, artificial intelligence, virtual and augmented reality, and more, and is not restricted to any particular platform [245]. The term “metaverse” refers to a

collective virtual shared space where users can interact in real-time with a computer-generated environment and other users [246]. In the metaverse, users can explore and interact with the virtual world, create personalized avatars, and communicate with other users in real-time.

A case study has been conducted in South African universities to explore the use of the metaverse for student examinations. The major benefits of using the metaverse in education include increased engagement and motivation, real-time feedback [247], personalized learning environments [248], access to a wide range of resources pertaining to virtual libraries, museums, and galleries [249], and collaborative experiences [250]. Additionally, immersive settings can improve student collaboration and communication. Students can interact with one another as avatars in virtual reality environments, which can overcome barriers and boost involvement. This is especially beneficial for students who may be hesitant to speak up in traditional classroom settings.

Limitations: The major challenges encountered by the students during the case study include readiness, hardware and software requirements, security and privacy concerns, and the role of faculty in ensuring a successful transition. They are explained in detail below:

1. **Student Readiness:** The major challenge faced by students during the examination is adapting to the new learning environment. Though the students are tech-savvy and adaptable to the virtual environment, they find it difficult to adapt to the new format of learning. A separate training session needs to be provided to the students so that they are well-versed in the metaverse learning environment.
2. **Hardware and Software:** Appropriate hardware and software are required, and they must be equipped with high-end graphics and processing capability.
3. **Security and Privacy:** Appropriate security measures must be adopted to protect student data against hacking, data breaches, and security assaults. Universities must regulate access and administrative controls to establish data protection protocols, ensuring student identity and data privacy.
4. **Role of Faculty:** Faculty should have a thorough understanding of using metaverse technology to benefit students during examinations and facilitate learning. It is the faculty's responsibility to ensure that the metaverse environment is suitable for students and aligns with the relevant syllabus and course materials.
5. **Accessibility:** All students, irrespective of their income backgrounds and learning capabilities, should have access to virtual desktop infrastructure. Students with disabilities must also have access to high-speed internet connectivity and powerful computing facilities.
6. **Equity:** Virtual settings, for instance, may not fairly accurately represent the diversity of racial, ethnic, or cultural backgrounds among students, which could cause them to feel excluded and disengaged. Additionally, disparities in educational performance may arise from unequal access to technological tools for students from underprivileged groups. To tackle these issues, universities should involve a diverse student body in designing virtual environments, ensure that the platforms are inclusive and culturally sensitive, and offer targeted support to students from underrepresented groups.

6 Valuable Insights

This section provides valuable insights from the literature survey conducted on online proctoring systems. The survey identifies research gaps in various areas, including head pose estimation, eye gaze estimation, face recognition, activity recognition, user verification, authentication, and security. By going through the survey, one can obtain a depth of knowledge on technological advancements, implementations, datasets, and other future technologies related to proctoring. The survey understands the importance of specific algorithms (like CNN, MTCNN, and FaceNet) versus general computer vision tasks and depicts them. Lastly, it is imperative to identify risks and counter measures to develop better full proof strategies to combat.

6.1 Eye gaze estimation

Eye gaze estimation solutions are often expensive, require manual intervention, and can be unreliable and inaccurate in practical deployments due to low image quality and variable lighting conditions. Eye gaze estimation has been broadly categorized into two types: model-based eye gaze estimations and appearance-based eye gaze models [251, 252]. Eye gaze-based models require a dedicated device, whereas appearance-based models do not [253]. Appearance-based models can be single-user or multi-user, with the latter facing challenges in unrestricted environments due to problems such as occlusion of the eyes, face, or body [254], wearing of spectacles [255], subject-to-camera distance [256], and various illumination variations [257]. Existing datasets also pose limitations for multi-user gaze estimation, and DL-based multi-user eye gaze estimation in unconstrained environments remains largely unexplored due to the lack of generic, publicly available datasets for this purpose [258].

Eye gaze estimations based on CNN suffer from problems such as head and eye rotations, which can be alleviated by using multiple cameras. Every CNN method encounters a prediction error greater than 0.5 due to an unbalanced dataset, variations in capturing data, and variations with respect to device, light, illumination, environment, size, and other external factors. Implementation of CNN is highly costly as it requires huge amounts of memory and computational ability. Heatmap-based models perform worse than 2D gaze estimation models as they are highly sensitive to hyperparameters and challenging to train [259, 260].

6.2 Head pose estimation

Head pose estimation is typically the reconstruction of the 3D head image from a 2D image [261]. It generally includes three angles: yaw, roll, and pitch. There are two categories of head pose estimation: model-based and appearance-based [262]. High accuracy is achieved by using facial landmark data for 3D reconstruction, which is part of the model-based approach. However, this model is highly dependent on the accuracy of the facial landmarks extracted [263]. Appearance-based models use the input as an image for feature extraction and calculations to produce outputs. Head poses change over time and can be modeled as a regression problem. However, the performance and accuracy of the head pose estimation models suffer due to limited generalizability and convergence. Additionally, camera calibration pertaining to real-world scenarios is a tedious and time-consuming process that needs to be addressed.

6.3 Facial recognition

Face recognition in proctoring involves both single and multiple face recognitions in front of the camera during the examinations. There is a high possibility that a student might cheat by using a still image in front of the camera [264]. In order to detect this, facial emotion recognition has been adopted, and eye blinking can be monitored to confirm the student's presence throughout the examination [265]. Facial recognition in proctoring is primarily used to ensure that the correct student is taking the exam. Variation in the facial poses is analyzed statistically using methods like eigen poses or eigen-faces [266, 267]. Continuous face biometric recognition systems such as PCA, Local Binary Pattern Histograms (LBPHs), KNNs, and CNNs are employed to ascertain the accuracy of the face recognition models [268, 269]. Visual attention guidance models are used to pinpoint visible areas on an obscured face, eliminating the need for additional parameters to predict the location and scale of the face [270].

6.4 Activity recognition

To maintain the integrity of the examination, it is important to look out for suspicious behaviors by students, such as looking around, making gestures, sharing answers, copying from notes, using electronic items, or leaving the exam hall without permission from the invigilator. Advanced deep learning algorithms are used to detect various human activities to aid in efficient automatic monitoring. However, activity recognition faces significant challenges such as occlusion, illumination, object size variations, differences in clothing and appearance, and computational time [271]. Activity recognition is difficult because it involves the complex classification of motion and gestures [272]. Even though CNN demonstrates the best performance, challenges related to similarities between different activities and objects being partially hidden still persist [273].

6.5 User verification

In the case of online proctoring systems, there is a risk of cheating as there is no interaction between the proctor and the student. Students can turn the camera toward video lectures. To address this, multiple biometrics are combined, such as face recognition with fingerprint scanning, fingerprint scanning with mouse usage patterns, or fingerprint scanning with vocal traits. Behavioral biometrics like keystrokes and mouse patterns offer continuous and transparent authentication, detecting any suspicious activity by students. This has led to the development of multi-biometric authentication systems that integrate facial and fingerprint recognition. These systems involve voice, touch, keystrokes, and mouse patterns, which are combined with facial recognition. However, there are challenges in human activity recognition, as collecting biometric data interrupts the student's explicit actions, which could be problematic if the student has limited time to complete the exam.

Fingerprint recognition is also affected by poor finger conditions. The use of face recognition to detect suspicious activity involves analyzing video frames to gain cues and traits for cheating. However, this demands significant storage, processing capabilities, and high bandwidth consumption. Temporal window-based solutions aim to extract information from facial images in a more streamlined manner, but behavioral biometrics may not be consistent over time. Most existing systems focus on the development of solutions that ensure the presence of the student rather than the interaction of the

student for user verification. Thus, hybrid algorithms are necessary to facilitate effective user verification in proctoring systems [274, 275, 276, 277, 278]. Optimization algorithms are also beneficial in improving the accuracy of student presence during examinations, thus enhancing the performance of user verification models [279].

6.6 Authentication and security

Privacy and security are the two core components that must be ensured. Primarily due to the complex processing involved in proctoring, security controls are highly concentrated, more than employing security protocols. Administrative controls, such as firewalls, access permissions, and technological controls, are highly preferred. Protocols like identity-based, certificate-based, certificate-less, and symmetric encryption techniques offer benefits in providing both security and privacy. Image-based hashing has yet to be concentrated for authentication in the case of online proctoring systems.

6.7 Heuristic adaptation of AI models

The models form the heart of AI working, in online proctoring, MTCNN detects faces, FaceNet verifies identity, and CNN-based models monitor suspicious behavior. These models are interlinked with each other for robust functioning.

MTCNN is a specialized CNN architecture designed specifically for face detection & alignment. It uses multi-task learning to both detect face bounding boxes and locate facial landmarks. MTCNN is described to function as the eyes of the proctoring system. MTCNN is one of the deployed vision AI services. Like most computer vision services, it runs in the cloud. MTCNN operates in three Cascade CNN stages, each of which performs a distinct function. The first stage is the Proposal stage, the second is the Refinement stage, and the last stage is the Output stage. Each of the cascade stages is responsible for capturing bounding boxes and the five landmarks, eyes, a nose, and the corners of a mouth. The MTCNN face recognition services in proctoring checks if a face is in the frame, if it is centered, and if it is facing the camera. MNNET immediately triggers an integrity flag if no face is detected (which implies the candidate has left the seat) or if more than one face is detected (which implies the candidate is getting assistance from another person).

FaceNet is a face recognition model, not a face detector. It converts the detected face into a numeral representation referred to as embedding (typically a 128 or 512 vector). It utilizes a Triplet Loss function during training to ensure that the distance between embeddings of the same person is minimized while the distance between different people is maximized. The image captured at the initial registration phase is cross examined with live exam feed; if the Euclidean distance exceeds a specific threshold, the system flags an “impersonation” attempt.

While MTCNN and FaceNet are specialized CNN architectures, a custom CNN is often deployed in proctoring to handle broader behavioral classification tasks beyond identity. This model is trained to recognize patterns such as object detection (identifying phones, books, or notes) and facial expression/emotion analysis (detecting unusual stress or eye-gaze shifts). By convolving through image frames to extract spatial hierarchies, CNN acts as the “logic layer” that categorizes specific movements like frequently looking away from the screen as suspicious behaviors that warrant a proctor’s review. CNN is therefore a building block, not a complete face-recognition system.

Table 20 Threats and countermeasures in AI-based proctoring

Threat category	Specific threats	Attack methods	potential impact	Countermeasures
Technical security threats				
Adversarial attacks	Evasion attacks	Training data manipulation; adversarial input patterns	False negatives	Train on adversarial examples; validate datasets
Spoofing	Face spoofing via photos/videos; deepfakes	Deepfakes	Bypass of Identity verification procedures	Liveness detection (multi-modal): Combine 3D depth sensing, micro-expression analysis, thermal imaging
System Vulnerabilities	Unauthorized access	Exploiting software flaws; network attacks	System compromise; data breach	Robustness testing: certificate-based authentication
Privacy threats				
Data breach	Unauthorized access to biometric data; fraud and misuse of data; collection of personal data	Inadequate encryption; data selling to third parties; secondary usage	Privacy violation; legal liability; GDPR violation; trust erosion; regulatory penalties	Collect only necessary data; time-bound data retention; access control; obtaining prior informed consent; zero-knowledge Encryption; Differential privacy
Algorithmic bias threats				
Facial recognition bias	Lower accuracy for people of color; women	Unrepresentative training datasets	Discriminatory outcomes; unfair flagging	Regular audits
Behavioral Bias	Cultural/disability differences flagged as suspicious	Unrepresentative training datasets	Accessibility violations; discrimination	Stakeholder consultation to factor in representation in intervention design
Environmental Bias	Penalizes students in poor lighting, noisy environments, infrastructure failures	Socioeconomic factors not included in intervention design; technical failures	Socioeconomic inequality and Equity violations	Pilot testing in representative settings
Human factor threats				
Psychological harm	Increased anxiety; test performance degradation	Constant surveillance; fear of false accusation	Mental health impacts	Equity impact assessments; pre-exam familiarization; robust appeals process; mental health support
Digital Divide	Exclusion of students lacking technology/bandwidth	Socioeconomic inequality	Educational inequity; access barriers	

6.8 Risk and mitigation strategies in AI-based proctoring

Rapid development of AI-based proctoring systems introduces a unique set of security challenges, ranging from sophisticated technical exploits to behavioral bypass attempts. A taxonomy of threats, and the corresponding mitigation strategies is detailed in Table 20.

7 Future research directions

Proctoring involves monitoring the examination and assisting the students. In the case of physical proctoring, the students who are involved in cheating/malpractice can be identified during the physical examination. However, this kind of physical proctoring is unable to catch the various types of cheating behavior that students engage in during

exams [280]. This poses the need for an online proctoring solution that effectively identifies the cheating behavior of the student. Due to the utilization of various technologies that have to run in the background to identify various forms of cheating behavior, the design of semi-automated and fully automated online proctoring systems has been complex. Problems pertaining to security, implementation, identity verification, administrative controls, data acquisition, data analysis, and data processing still persist. This section provides deep, valuable insights into the future research directions of each of the application requirements corresponding to online proctoring systems.

- Implementation of CNNs for eye gaze estimation suffers from problems related to generalization due to light, illumination, data capture, devices, eyeglasses, camera specifications, etc. [281, 282]. This can be eradicated when the dataset becomes huge and involves a trivial collection of vast datasets with various parameters involving eye, face, body, and head movements. In order to track the eye's gazes, an appropriate gaze-aware UI element has to be developed. Multi-user and Multi-camera-based gaze estimation models and 3D appearance-based gaze estimation models can provide benefits in the future. The development and design of a CNN to cater to the need for generalization can boost the performance of the network.
- Location-guided head pose estimation is a new arena where major challenges are yet to be addressed [283].
- Blockchain-based, biometric-based, and privacy-preserving authentication schemes have to be developed in order to ensure authentication and security. Blockchain offers a Ubiquitous Learning Environment (ULE), thereby providing hard and tamper-proof security for sensitive information [284]. Blockchain offers decentralized, immutable platforms [285, 286], high storage capability [287], identity management [288], transactions and certificates [289, 290] with reduced cost [291], and translucent access [292]. Blockchain also offers a Learning Logs (BOLL) platform for the storage of learning records and a Smart Ecosystem for Learning and Inclusion (SELI) [293, 294], revolutionizing proctoring to harness new dimensions in providing security.
- Hyperautomation/Robotic Process Automation (RPA) provides efficient data cleansing, proctoring, intelligent document processing, query processing, order processing, and automated report generation services, resulting in the development of autonomous systems for proctoring [295, 296].
- Internet of Things (IoT) with information and communication technology provides various capabilities to monitor, proctor, observe, listen, and perform different tasks such as body movements, facial expressions, time taken for answering a question, personalizing, and dynamic customization of various teaching and assessment capabilities [297, 298].
- Cloud computing can be harnessed to support various proctoring tasks through services such as SaaS, PaaS, DaaS, PaaS, PaaS, and IaaS. This allows activities to be carried out by enriching a shared database and storage, thereby achieving virtualization. This provides the autonomous functioning of proctoring systems, thereby increasing the availability of CPUs, memory, and bandwidth for collaboration [299].
- Extended Reality (XR) is feasible with the help of VR, MR, and AR in education and proctoring. Based on 3D modeling and 360-degree photos, Virtual Reality (VR)

offers simulated experiences outside of space and time. Augmented Reality (AR) adds computer-generated elements to an existing reality, such as games, audio, 3D models, photos, videos, graphics, animated sequences, and GPS data, to make it seem more realistic [300, 301]. Mixed Reality provides immersive realism by coagulating AR and VR. These kinds of applications will benefit fully automated proctoring systems in the near future.

- Artificial intelligence-based systems are harnessed to recognize gestures; intelligent tutoring, learning, and proctoring, voice assistants, and bots with AI capabilities [302, 303] for proctoring can be adapted.
- Significant ethical and social challenges exist in this study domain that must be systematically analyzed. These concerns primarily center on student privacy, algorithmic bias, and the psychological impact of constant surveillance. The major areas to be discovered include privacy and data sovereignty, algorithmic bias and digital exclusion, psychological impact and test anxiety. All the above areas must be researched keeping in view the ethical, social, psychological, and legal dimension.

8 Conclusion

In this survey, different proctoring systems have been classified and discussed. Online proctoring systems are broadly classified into six major types: eye gaze tracking, head pose estimation, activity recognition, facial recognition, user verification, and authentication and security. A comparative analysis has been provided in terms of problems addressed, and key features, benefits, and limitations related to various features utilized in proctoring systems are compared. The survey also provides valuable insights and potential research gaps, along with the commercially available proctoring software, which is also analyzed and reviewed. Further future directions, along with conceptual scenarios, are presented to ascertain the effectiveness of the proposed survey, which will aid future researchers.

Alternative technologies such as blockchain technology, cloud computing, big data, extended reality, and security protocols are considered for designing appropriate architecture for online proctoring systems. Moreover, investigating Metaverse, Avatar, and Omni-Sensing Technology (OST) will be the new milestone that drives the future of online proctoring systems.

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Author contributions

M.M. and I.C. jointly conceived the research idea and developed the methodology for the study. M.M. contributed to conceptualization, formal analysis, methodology, data curation, and writing the original draft. I. C. took the responsibility of review, editing, supervision, and project administration.

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No datasets were generated or analysed during the current study.

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